

The Bright Side of Mathematics

The following pages cover the whole Ordinary Differential Equations course of the Bright Side of Mathematics. Please note that the creator lives from generous supporters and would be very happy about a donation. See more here: <https://tbsom.de/support>

Have fun learning mathematics!

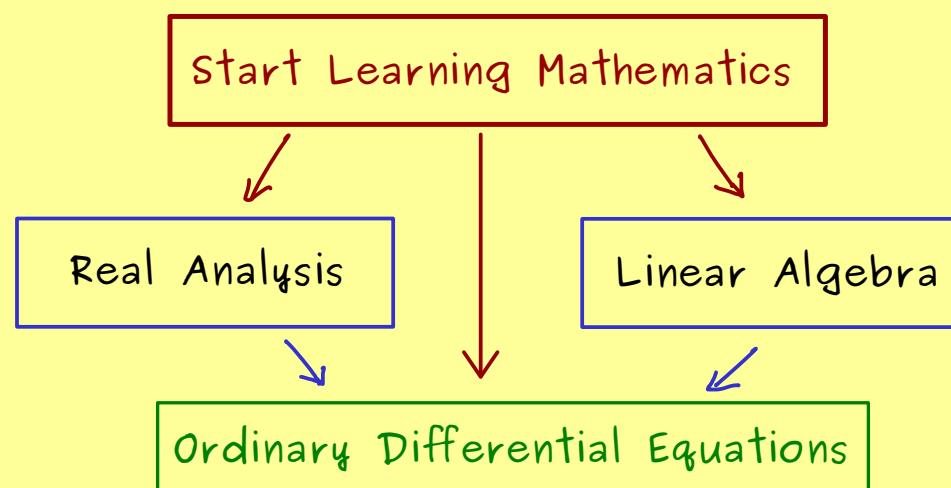


Ordinary Differential Equations – Part 1

$$f' = f \leftarrow$$

search for a function f that satisfies this?

$$f(x) = e^x$$



Other examples: (a) $\ddot{x} = -\omega^2 x$ (harmonic oscillator) (second order derivatives)

(b)

$$\left. \begin{aligned} m \cdot \ddot{x} &= F \\ m \cdot \ddot{y} &= F \\ m \cdot \ddot{z} &= F \end{aligned} \right\} \text{system of differential equations}$$

- Topics:
- system of ordinary differential equations (ODE)
 - solution methods
 - existence and uniqueness of solutions
 - linear ordinary differential equations (matrix exponential function)



Ordinary Differential Equations – Part 2

Definitions: For $I \subseteq \mathbb{R}$ (interval, open set, union intervals,...)

$$C^k(I) := \left\{ x: I \rightarrow \mathbb{R} \mid \begin{array}{l} x \text{ is } k\text{-times continuously differentiable} \\ t \mapsto x(t) \end{array} \right\}$$

$\dot{x}, \ddot{x}, \dots, x^{(k)}$ continuous functions
 $x' = \frac{dx}{dt}$

Ordinary differential equation: $F(t, x, \dot{x}, \dots, x^{(k)}) = 0$

→ ODE

continuous

Example: $t + x + 2\dot{x} + (\ddot{x})^2 = 0$

(explicit) ODE of order 1: $\dot{x} = w(t, x), w: I \times J \rightarrow \mathbb{R}, I, J \subseteq \mathbb{R}$
intervals

Example: $\dot{x} = x + t$

What about? $\begin{pmatrix} \dot{x}_1 = x_2 + t \\ \dot{x}_2 = x_1 + t \end{pmatrix} \rightsquigarrow \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = w\left(t, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)$

System of (explicit) ODEs of order 1:

$$\dot{x} = w(t, x), \quad x(t) \in \mathbb{R}^n, \quad w: I \times U \rightarrow \mathbb{R}^n$$

↑ open set in $\mathbb{R}^n$

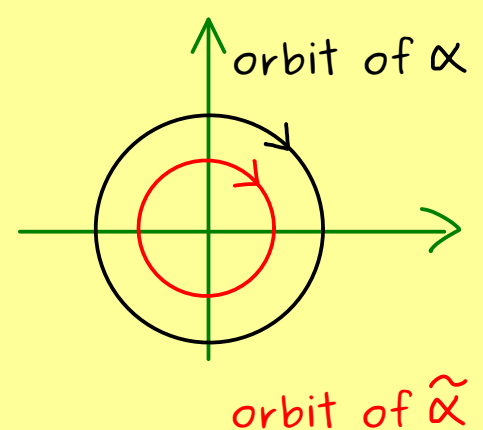
Solution of ODE: $\alpha: (t_0, t_1) \rightarrow U$ with $(t_0, t_1) \subseteq I$

satisfies $\dot{\alpha}(t) = w(t, \alpha(t))$ for all $t \in (t_0, t_1)$.

Example: $\begin{array}{l} \dot{x}_1 = x_2 \\ \dot{x}_2 = -x_1 \end{array}, \quad n=2, \quad U = \mathbb{R}^2, \quad w(t, x) = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix}$

$\rightsquigarrow \alpha(t) = \begin{pmatrix} \sin(t) \\ \cos(t) \end{pmatrix}$ is a solution

$\tilde{\alpha}(t) = \frac{1}{2} \begin{pmatrix} \sin(t) \\ \cos(t) \end{pmatrix}$ is a solution





Ordinary Differential Equations – Part 3

ODE: $\dot{x} = w(t, x)$ (explicit, of first order)

Example: (a) $\dot{x} = \lambda \cdot x \rightsquigarrow$ autonomous

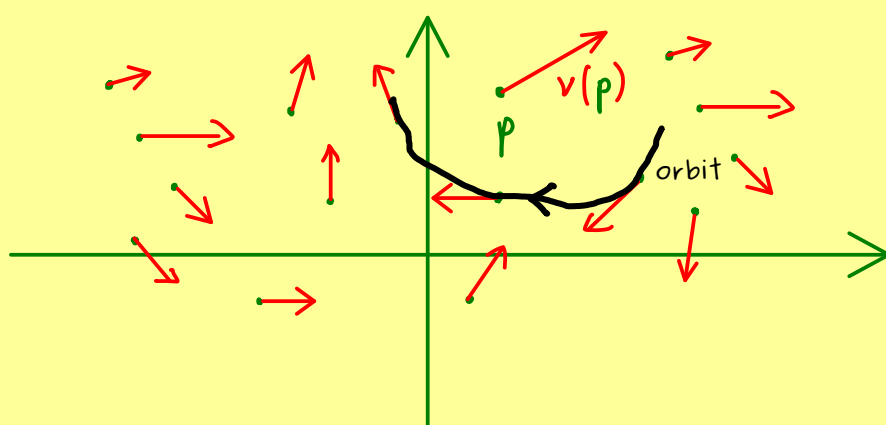
(b) $\dot{x} = t \rightsquigarrow$ not autonomous

(c) $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix} \rightsquigarrow$ autonomous

Definition: autonomous system: $\dot{x} = v(x)$ with $v: U \rightarrow \mathbb{R}^n$
 $\swarrow$
 $U \subseteq \mathbb{R}^n$

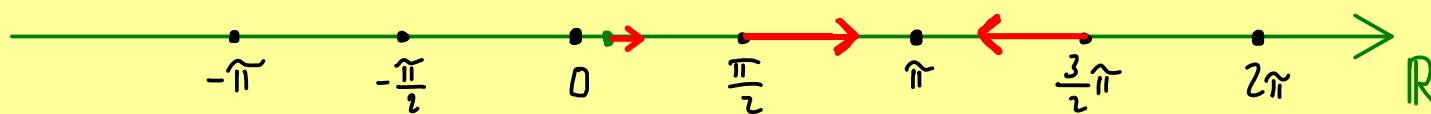
often:
 U open
 v continuous

Directional field:



$$v: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

Examples: (a) $\dot{x} = \sin(x)$, $v: \mathbb{R} \rightarrow \mathbb{R}$, $v(x) = \sin(x)$

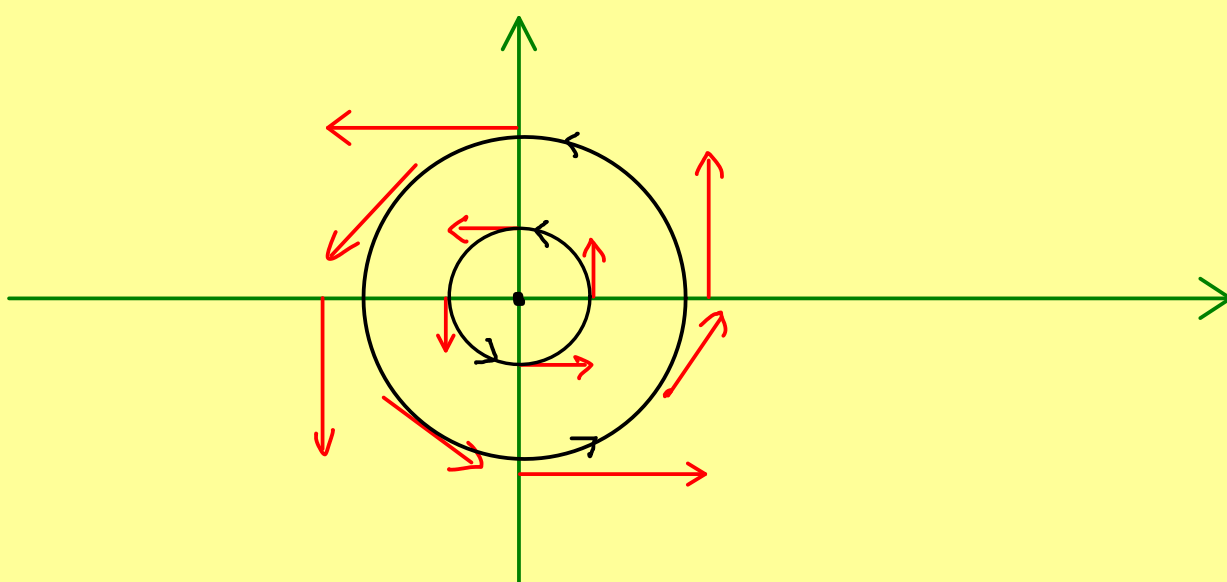


(1) $\alpha(t) = 0$ for all $t \in \mathbb{R}$ is a solution: $\underbrace{\dot{\alpha}(t)}_{=0} = \underbrace{\sin(\alpha(t))}_{=0}$

(2) $\alpha(t) = \pi$ for all $t \in \mathbb{R}$ is a solution.

(3) A solution with $\alpha(0) = \frac{\pi}{2}$ is monotonically increasing
 with $\lim_{t \rightarrow \infty} \alpha(t) = \pi$.

(b) $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}$, $v: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $(x_1, x_2) \mapsto \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}$





Ordinary Differential Equations – Part 4

Example: $\ddot{x} = \cos(\ddot{x}) + \dot{x}^2 + x$ (autonomous ODE of third order)

define: $y = \begin{pmatrix} x \\ \dot{x} \\ \ddot{x} \end{pmatrix} \rightsquigarrow$

$$\dot{y}_1 = y_2$$

$$\dot{y}_2 = y_3$$

$$\dot{y}_3 = \cos(y_3) + y_2^2 + y_1$$

$$\Rightarrow \dot{y} = v(y) \quad (\text{autonomous system of ODEs of first order})$$

Example: $\ddot{x} = \cos(\ddot{x}) + \dot{x}^2 + x - t^4$ (non-autonomous ODE of third order)

define: $\begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} t \\ x \\ \dot{x} \\ \ddot{x} \end{pmatrix} \rightsquigarrow$

$$\dot{y}_0 = 1$$

$$\dot{y}_1 = y_2$$

$$\dot{y}_2 = y_3$$

$$\dot{y}_3 = \cos(y_3) + y_2^2 + y_1 - y_0^4$$

Remember: (explicit) autonomous ODE of n th order $\Leftrightarrow \dot{y} = v(y)$

$\nwarrow$ n components

(autonomous system of n ODEs of first order)

(explicit) non-autonomous ODE of n th order $\Leftrightarrow \dot{y} = v(y)$

$\nwarrow$ $n+1$ components

(autonomous system of $n+1$ ODEs of first order)



Ordinary Differential Equations – Part 5

Initial value problem: $\dot{x} = v(x)$ with $v: \mathbb{R} \rightarrow \mathbb{R}$ continuous

$$x(0) = x_0$$

Find all solutions $\alpha: (t_0, t_1) \rightarrow \mathbb{R}$ ($\dot{\alpha}(t) = v(\alpha(t))$)
with $\alpha(0) = x_0$

Solving strategy: Assume $v(x_0) \neq 0$:

$$\text{ODE: } \frac{\dot{x}}{v(x)} = 1$$

Therefore: any solution $\alpha: (t_0, t_1) \rightarrow \mathbb{R}$ with $\alpha(0) = x_0$ satisfies:

fundamental
theorem
of calculus

$$\frac{\dot{\alpha}(s)}{v(\alpha(s))} = 1 \quad \text{for all } s \in (t_0, t_1)$$

$$\Leftrightarrow \int_0^t \frac{\dot{\alpha}(s)}{v(\alpha(s))} ds = t \quad \text{for all } t \in (t_0, t_1)$$

substitution: $x = \alpha(s)$, $dx = \dot{\alpha}(s) ds$

$$\Leftrightarrow \int_{x_0}^{\alpha(t)} \frac{1}{v(x)} dx = t \quad \text{for all } t \in (t_0, t_1)$$

$$\Leftrightarrow F(\alpha(t)) - F(x_0) = t \quad \text{for all } t \in (t_0, t_1)$$

where F is an antiderivative of $\frac{1}{v}$

$$\Leftrightarrow F(\alpha(t)) = t - c \quad \text{for all } t \in (t_0, t_1)$$

$$\Leftrightarrow \alpha(t) = F^{-1}(t - c) \quad \text{for all } t \in (t_0, t_1)$$

Examples:

(a) $\dot{x} = \lambda \cdot x$, $x(0) = x_0 \neq 0$

$$\Leftrightarrow \frac{dx}{dt} = \lambda \cdot x \quad \Leftrightarrow \int \frac{dx}{x} = \int \lambda dt \quad \text{informally}$$

$$\Leftrightarrow \log(|x|) = \lambda \cdot t + C, \quad C \in \mathbb{R}$$

natural logarithm

$$\Leftrightarrow |\alpha(t)| = e^{\lambda t} \cdot e^C$$

$$\Leftrightarrow \alpha(t) = \begin{cases} -e^C e^{\lambda t} \\ e^C e^{\lambda t} \end{cases}$$

$$\text{solution: } \alpha(t) = x_0 \cdot e^{\lambda t}$$

(b) $\dot{x} = x^2$, $x(0) = x_0 \neq 0$

$$\Leftrightarrow \frac{dx}{dt} = x^2 \quad \Leftrightarrow \int \frac{dx}{x^2} = \int dt$$

$$\Leftrightarrow -\frac{1}{x} = t + C, \quad C \in \mathbb{R}$$

$$\Leftrightarrow -\frac{1}{\alpha(t)} = t + C, \quad C \in \mathbb{R}$$

$$\Leftrightarrow \alpha(t) = \frac{-1}{t + C}, \quad C \in \mathbb{R}$$

$$\text{initial value: } \alpha(0) = \frac{-1}{C} \stackrel{!}{=} x_0 \Rightarrow \boxed{C = -\frac{1}{x_0}}$$

$$\text{solution: } \alpha(t) = \frac{-1}{t + (-\frac{1}{x_0})} = \frac{x_0}{1 - x_0 t}$$



Ordinary Differential Equations – Part 6

non-autonomous ODE: $\dot{x} = w(t, x)$ can we separate t and x ?

example:

$$\dot{x} = \underbrace{t^3}_{\text{only } t} \cdot \underbrace{x^2}_{\text{only } x}$$

Separation of variables:

$$\dot{x} = g(t) \cdot h(x), \quad x(t_0) = x_0 \quad (\text{initial value problem})$$

continuous functions

$$\text{Assume: } h(x_0) \neq 0 \Rightarrow \frac{\dot{x}}{h(x)} = g(t)$$

Therefore: any solution $\alpha: (t_1, t_2) \rightarrow \mathbb{R}$ with $\alpha(t_0) = x_0$ satisfies:

fundamental
theorem
of calculus

$$\frac{\dot{\alpha}(s)}{h(\alpha(s))} = g(s) \quad \text{for all } s \in (t_1, t_2)$$

$$\Leftrightarrow \int_{t_0}^t \frac{\dot{\alpha}(s)}{h(\alpha(s))} ds = \int_{t_0}^t g(s) ds \quad \text{for all } t \in (t_1, t_2)$$

substitution: $x = \alpha(s)$, $dx = \dot{\alpha}(s) ds$

$$\Leftrightarrow \int_{x_0}^{\alpha(t)} \frac{1}{h(x)} dx = \int_{t_0}^t g(s) ds \quad \text{for all } t \in (t_1, t_2)$$

$$\Leftrightarrow F(\alpha(t)) - F(x_0) = G(t) - G(t_0) \quad \text{for all } t \in (t_1, t_2)$$

where F is an antiderivative of $\frac{1}{h}$

where G is an antiderivative of g

$$\Leftrightarrow F(\alpha(t)) = G(t) + c$$

for a constant $c \in \mathbb{R}$, for all $t \in (t_1, t_2)$

$$\Leftrightarrow \alpha(t) = F^{-1}(G(t) + c)$$

Example:

(a) $\dot{x} = \frac{1}{3}t^3 x$, $x(0) = x_0 \neq 0$

$$\Leftrightarrow \frac{dx}{dt} = \frac{1}{3}t^3 x \quad \xRightarrow{\text{informally}} \int \frac{dx}{x} = \int \frac{1}{3}t^3 dt$$

$$\Leftrightarrow \log(|x|) = \frac{1}{12}t^4 + c \quad \text{for a constant } c \in \mathbb{R}$$

natural logarithm

$$\Leftrightarrow |\alpha(t)| = e^{\frac{1}{12}t^4 + c} \quad \xRightarrow{\alpha(0) = x_0} \alpha(t) = x_0 \cdot e^{\frac{1}{12}t^4}$$

(b) $\dot{x} = \sin(t) \cdot e^x$, $x(0) = x_0$

$$\Leftrightarrow \frac{dx}{dt} = \sin(t) \cdot e^x \quad \xRightarrow{\text{informally}} \int \frac{dx}{e^x} = \int \sin(t) dt$$

$$\Leftrightarrow -e^{-x} = -\cos(t) + c \quad \text{for a constant } c \in \mathbb{R}$$

$$\Leftrightarrow \alpha(t) = -\log(\cos(t) + \tilde{c}) \quad \text{for a constant } \tilde{c} \in \mathbb{R}$$

$$\xRightarrow{\alpha(0) = x_0} -\log(\cos(0) + \tilde{c}) = x_0 \Rightarrow \tilde{c} = e^{-x_0} - 1$$



Ordinary Differential Equations – Part 7

Linear ODE of first order: $\dot{x} = a(t) \cdot x + b(t)$ continuous functions

Finding solutions: (with an integrating factor)

$$\dot{x} + \tilde{a}(t)x = b(t)$$

with $\tilde{a}(t) := -a(t)$

multiplying both sides

$$\Leftrightarrow \dot{x} e^{\tilde{A}(t)} + \tilde{a}(t)x e^{\tilde{A}(t)} = b(t)e^{\tilde{A}(t)}$$

product rule

$$\Leftrightarrow \frac{d}{dt} \left(x(t) e^{\tilde{A}(t)} \right) = b(t) e^{\tilde{A}(t)}$$

antiderivative

$$\Leftrightarrow x(t) e^{\tilde{A}(t)} = H(t) + c, \quad c \in \mathbb{R}$$

$$\text{solutions: } \alpha(t) = e^{-\tilde{A}(t)} (H(t) + c), \quad c \in \mathbb{R}$$

Note: if $\tilde{A}$ is an antiderivative of $\tilde{a}$,

$$\text{then: } \frac{d}{dt} e^{\tilde{A}(t)} = \underbrace{\tilde{A}'(t)}_{\tilde{a}(t)} \cdot e^{\tilde{A}(t)}$$

H is antiderivative of $b(t)e^{\tilde{A}(t)}$

Example: $\dot{x} = tx + e^{\frac{1}{2}t^2}, \quad x(0) = x_0$

$$\Leftrightarrow \dot{x} - tx = e^{\frac{1}{2}t^2} \quad | \cdot e^{-\frac{1}{2}t^2}$$

$$\Leftrightarrow \dot{x} \cdot e^{-\frac{1}{2}t^2} - tx e^{-\frac{1}{2}t^2} = 1$$

$$\Leftrightarrow \frac{d}{dt} \left(x(t) \cdot e^{-\frac{1}{2}t^2} \right) = 1$$

$$\Leftrightarrow x(t) \cdot e^{-\frac{1}{2}t^2} = t + c, \quad c \in \mathbb{R}$$

$$\Leftrightarrow \text{solution: } \alpha(t) = (t + c) \cdot e^{\frac{1}{2}t^2}$$

Initial value condition: $\alpha(0) = x_0 \rightsquigarrow \alpha(t) = (t + x_0) \cdot e^{\frac{1}{2}t^2}$
//
c



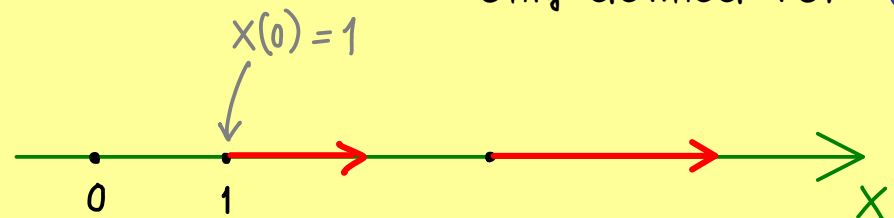
Ordinary Differential Equations – Part 8

Questions: Initial value problem: $\dot{x} = v(x)$ with $v: \mathbb{R}^n \rightarrow \mathbb{R}^n$ continuous
 $x(0) = x_0$

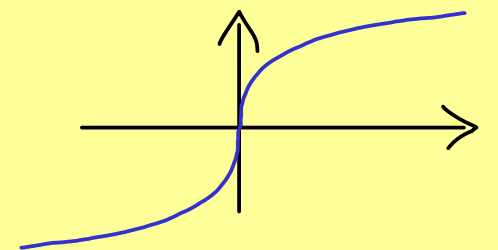
- Does a solution exist?
- What is the domain of definition?
- Uniqueness of solutions?

Examples: (a) $\dot{x} = x^2$, $x(0) = 1$ $\xRightarrow{\text{part 5}}$ solution exists: $\alpha(t) = \frac{1}{1-t}$

only defined for $t < 1$



(b) $\dot{x} = v(x)$, $x(0) = 0$ with $v(x) = \begin{cases} \sqrt{|x|}, & x \geq 0 \\ -\sqrt{|x|}, & x < 0 \end{cases}$

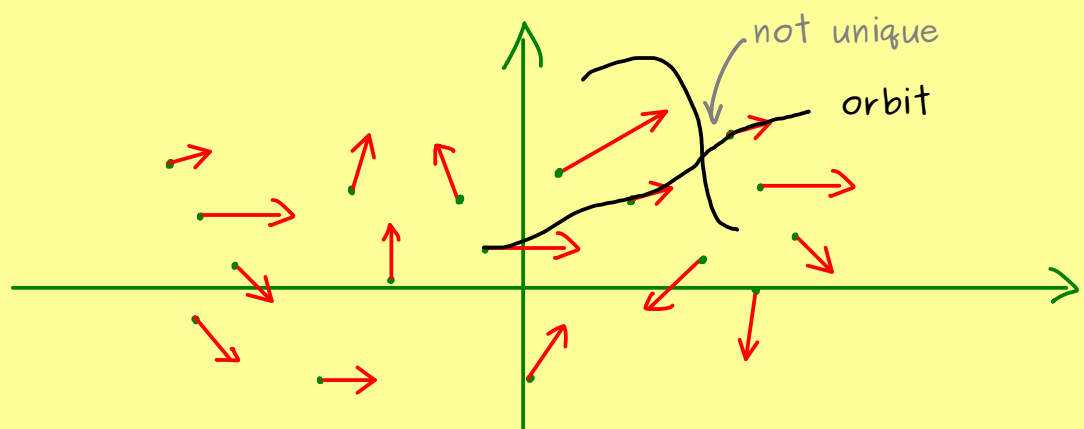


We find at least two solutions: $\alpha(t) = 0$ for all t

$$\tilde{\alpha}(t) = \begin{cases} 0, & t \leq 0 \\ \frac{1}{4}t^2, & t > 0 \end{cases}$$

In general:

directional field



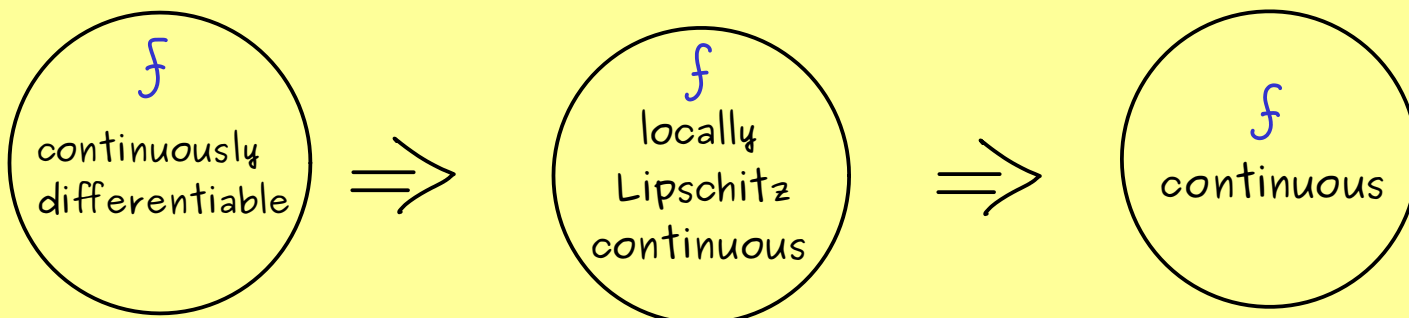
existence: does each point have an orbit?

uniqueness: can two orbits cross?

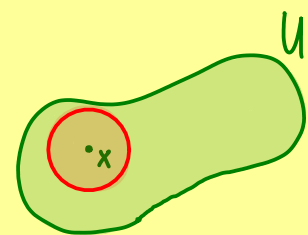


Ordinary Differential Equations – Part 9

$$f: \mathbb{R} \rightarrow \mathbb{R}$$



Definition: $v: \mathbb{R}^n \rightarrow \mathbb{R}^n$ (or open set U) is called locally Lipschitz continuous if:



$$\forall x \in \mathbb{R}^n \quad \exists \epsilon > 0 \quad \exists L \geq 0 \quad \forall y, z \in B_\epsilon(x) :$$

$$\|v(y) - v(z)\| \leq L \cdot \|y - z\|$$

$\swarrow$ Lipschitz constant
 $\nwarrow$ standard norm of $\mathbb{R}^n$

Remember: (1) v loc. Lipschitz continuous $\Rightarrow v$ continuous
 $\left(y_n \xrightarrow{n \rightarrow \infty} y \Rightarrow \|v(y_n) - v(y)\| \xrightarrow{n \rightarrow \infty} 0 \right)$

$$(2) \quad v \text{ loc. Lipschitz continuous} \Rightarrow \frac{\|v(y) - v(z)\|}{\|y - z\|} \leq L$$

(3) $f: \mathbb{R} \rightarrow \mathbb{R}$ continuously differentiable. Fix $x \in \mathbb{R}$, $\epsilon > 0$

$$\frac{|f(y) - f(z)|}{|y - z|} \stackrel{\substack{\text{mean} \\ \text{value} \\ \text{theorem}}}{=} |f'(\xi)| \quad \xi \text{ between } y \text{ and } z$$

$$\leq \sup_{\tilde{\xi} \in B_\epsilon(x)} |f'(\tilde{\xi})| =: L \geq 0$$

$\Rightarrow f$ loc. Lipschitz continuous



Ordinary Differential Equations – Part 10

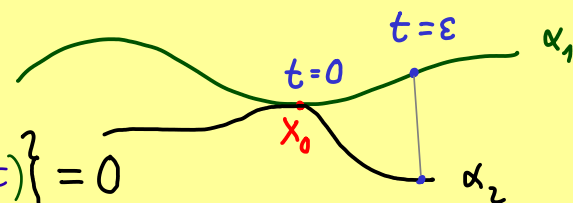
Initial value problem: $\dot{x} = v(x)$ with $v: \mathbb{R}^n \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous
 $x(0) = x_0$

Theorem: The initial value problem has at most one solution. (orbits don't cross!)

Proof: Assume α_1, α_2 are two distinct solutions $(\alpha_1(0) = \alpha_2(0) = x_0)$

with $\alpha_1(\varepsilon) \neq \alpha_2(\varepsilon)$ for $\varepsilon > 0$ and

$$\inf \{ \tau \in [0, \varepsilon] \mid \alpha_1(\tau) \neq \alpha_2(\tau) \} = 0$$



$$\|\beta(t)\| = \|\alpha_1(t) - \alpha_2(t)\|$$

$$= \left\| \int_0^t \dot{\alpha}_1(\tau) d\tau - \int_0^t \dot{\alpha}_2(\tau) d\tau \right\| = \left\| \int_0^t v(\alpha_1(\tau)) d\tau - \int_0^t v(\alpha_2(\tau)) d\tau \right\|$$

$$= \left\| \int_0^t (v(\alpha_1(\tau)) - v(\alpha_2(\tau))) d\tau \right\| \leq \int_0^t \|v(\alpha_1(\tau)) - v(\alpha_2(\tau))\| d\tau$$

$$\leq L \cdot \|\alpha_1(\tau) - \alpha_2(\tau)\|$$

$$\leq L \cdot \int_0^\varepsilon \|\beta(\tau)\| d\tau \leq L \cdot \varepsilon \cdot \max_{\tau \in [0, \varepsilon]} \|\beta(\tau)\|$$

choose ε such that $L \varepsilon \leq \frac{1}{2}$

$$\Rightarrow \boxed{\|\beta(t)\| \leq \frac{1}{2} \cdot \max_{\tau \in [0, \varepsilon]} \|\beta(\tau)\| \text{ for all } t \in (0, \varepsilon]} \Rightarrow \alpha_1 = \alpha_2 \quad \checkmark$$



Ordinary Differential Equations – Part 11

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases} \quad \begin{array}{l} \text{initial value problem} \\ \text{with } v: \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ loc. Lipschitz continuous} \end{array}$$

integrating $\rightarrow$

$$\underbrace{\int_0^t \dot{x}(s) ds}_{x(t) - x(0)} = \int_0^t v(x(s)) ds$$

$$\Rightarrow x(t) = x_0 + \underbrace{\int_0^t v(x(s)) ds}_{\Phi(x)} \quad \text{function}$$

Definition: $\Phi: \mathcal{F}(\mathbb{R}, \mathbb{R}^n) \rightarrow \mathcal{F}(\mathbb{R}, \mathbb{R}^n)$

$$f \mapsto \left(t \mapsto x_0 + \int_0^t v(f(s)) ds \right)$$

Now we know: $x: \mathbb{R} \rightarrow \mathbb{R}^n$ is a solution of

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases}$$

$$\Leftrightarrow \Phi(x) = x \quad (\text{fixed point equation})$$

Banach fixed-point theorem: Let (X, d) be a complete metric space (set with distance function)

and $\Phi: X \rightarrow X$ be a contraction, which means:

$$\exists q \in [0, 1) \quad \forall x, \tilde{x} \in X: d(\Phi(x), \Phi(\tilde{x})) \leq q \cdot d(x, \tilde{x}) \quad .$$

$\nwarrow < 1$

Then: Φ has a unique fixed point $x^* \in X$ and

for each $x_0 \in X$ we have: $\Phi^n(x_0) \xrightarrow{n \rightarrow \infty} x^*$.



Ordinary Differential Equations – Part 12

initial value problem:

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases}$$

with $v: \mathbb{R}^n \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous

$\Rightarrow$ there is a unique solution!

(Picard-Lindelöf theorem)

Banach fixed-point theorem:

Let (X, d) be a complete metric space

and $\Phi: X \rightarrow X$ be a contraction.

Then: Φ has a unique fixed point $x^* \in X$.

We need:

(1) Complete metric space consisting of functions.

(2) Contraction $\Phi(\alpha)(t) = x_0 + \int_0^t v(\alpha(s)) ds$

Now we know: $\alpha: \mathbb{R} \rightarrow \mathbb{R}^n$ is a solution of $\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases}$

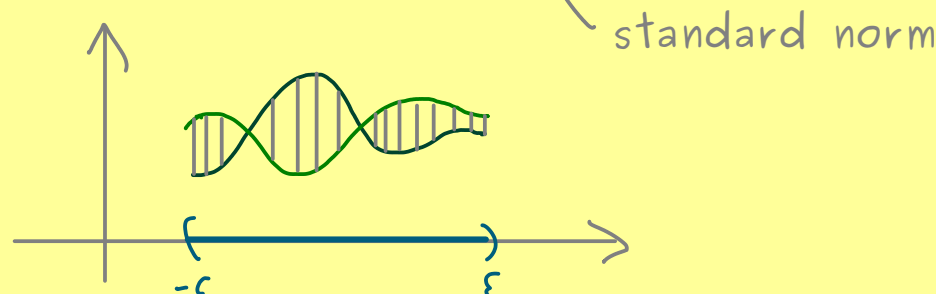
$$\Leftrightarrow \Phi(\alpha) = \alpha \quad (\text{fixed point equation})$$

For (1):

$$X = \left\{ \alpha: (-\varepsilon, \varepsilon) \rightarrow \tilde{U} \subseteq \mathbb{R}^n \mid \begin{array}{l} \text{in the domain of } v \\ \text{with property } (*) \\ \text{(see below)} \end{array} \mid \alpha \text{ continuous, } \alpha(0) = x_0 \right\}$$

with metric:

$$d(\alpha, \beta) := \sup_{t \in (-\varepsilon, \varepsilon)} \|\alpha(t) - \beta(t)\|_{\mathbb{R}^n}$$



Fact: (X, d) is a complete metric space.

For (2):

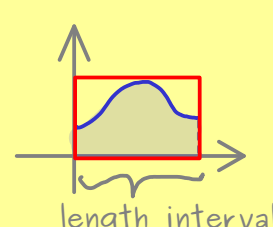
$$\Phi(\alpha)(t) = x_0 + \int_0^t v(\alpha(s)) ds \quad \text{gives a map } \Phi: X \rightarrow X$$

$$d(\Phi(\alpha), \Phi(\beta)) = \sup_{t \in (-\varepsilon, \varepsilon)} \|\Phi(\alpha)(t) - \Phi(\beta)(t)\|_{\mathbb{R}^n}$$

triangle inequality
for integrals

$$= \sup_{t \in (-\varepsilon, \varepsilon)} \left\| \int_0^t (v(\alpha(s)) - v(\beta(s))) ds \right\|_{\mathbb{R}^n}$$

$$\leq \sup_{t \in (-\varepsilon, \varepsilon)} \int_0^t \|v(\alpha(s)) - v(\beta(s))\|_{\mathbb{R}^n} ds$$



$$\leq \sup_{t \in (-\varepsilon, \varepsilon)} \underbrace{\text{length}([0, t])}_{|t| \leq \varepsilon} \cdot \sup_{s \in [0, t]} \|v(\alpha(s)) - v(\beta(s))\|_{\mathbb{R}^n} \leq \sup_{s \in (-\varepsilon, \varepsilon)} \dots$$

$$\leq \varepsilon \cdot \sup_{s \in (-\varepsilon, \varepsilon)} \|v(\alpha(s)) - v(\beta(s))\|_{\mathbb{R}^n} \leq L \|\alpha(s) - \beta(s)\|_{\mathbb{R}^n}$$

(*) needed

$$\leq \underbrace{\varepsilon \cdot L}_{< 1 \text{ for } \varepsilon \text{ small enough}} \cdot d(\alpha, \beta) \quad \text{contraction}$$

Picard-Lindelöf theorem

$v: U \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous, $x_0 \in U$.

Then there is $\varepsilon > 0$ and a unique solution $\alpha: (-\varepsilon, \varepsilon) \rightarrow U$

for the initial value problem

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases}$$

Definition of $\tilde{U}$ with property (*)

v being locally Lipschitz continuous at x_0 means:

$$\exists \delta > 0 \quad \exists L \geq 0 \quad \forall y, z \in \mathcal{B}_\delta(x_0) : \begin{array}{c} \|v(y) - v(z)\| \\ \uparrow \quad \uparrow \\ \alpha(s) \quad \beta(s) \end{array} \leq L \cdot \|y - z\|$$

So we need $\alpha(s), \beta(s) \in \mathcal{B}_\delta(x_0)$ for all $s \in (-\varepsilon, \varepsilon)$.

Hence: $\tilde{U} := \mathcal{B}_\delta(x_0)$ (not a problem for the solution since we choose ε as small as we want)



Ordinary Differential Equations – Part 13

Picard-Lindelöf theorem

$v: U \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous, $x_0 \in U$.

Then there is $\epsilon > 0$ and a unique solution $\alpha: (-\epsilon, \epsilon) \rightarrow U$

for the initial value problem

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases}.$$

Picard iteration:

Iteration from the Banach fixed-point theorem $\Phi^n(\tilde{\alpha}) \xrightarrow{n \rightarrow \infty} \alpha$

$$\Phi(\tilde{\alpha})(t) = x_0 + \int_0^t v(\tilde{\alpha}(s)) ds$$

Example: initial value problem: $\dot{x} = x$
 $x(0) = 1$

start with $\tilde{\alpha}: (-\epsilon, \epsilon) \rightarrow \mathbb{R}$, $\tilde{\alpha}(t) = 1$

first step: $\Phi(\tilde{\alpha})(t) = 1 + \int_0^t \tilde{\alpha}(s) ds = 1 + t$

second step: $\Phi^2(\tilde{\alpha})(t) = 1 + \int_0^t (1+s) ds = 1 + t + \frac{1}{2}t^2$

n th step: $\Phi^n(\tilde{\alpha})(t) = 1 + t + \frac{1}{2}t^2 + \frac{1}{6}t^3 + \dots + \frac{1}{n!}t^n$

$\downarrow n \rightarrow \infty$ (pointwise limit) (also uniform limit)

$$\sum_{k=0}^{\infty} \frac{t^k}{k!} = \exp(t)$$



Ordinary Differential Equations – Part 14

initial value problem:

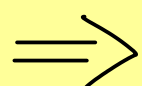
(IVP $_{x_0}^{t_0}$)

$$\dot{x} = v(x)$$

$$x(t_0) = x_0$$

with $v: \mathbb{D} \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous
 $\uparrow$ open in $\mathbb{R}^n$

(Picard-Lindelöf theorem)



there is $\varepsilon > 0$ and a unique solution

$$\alpha: (t_0 - \varepsilon, t_0 + \varepsilon) \rightarrow \mathbb{D}$$

Extension of solution:

We say a solution $\tilde{\alpha}: I \rightarrow \mathbb{D}$ extends $\alpha: (t_0 - \varepsilon, t_0 + \varepsilon) \rightarrow \mathbb{D}$ if $I \not\supseteq (t_0 - \varepsilon, t_0 + \varepsilon)$ and $\tilde{\alpha}|_{(t_0 - \varepsilon, t_0 + \varepsilon)} = \alpha$.

Maximal solutions:

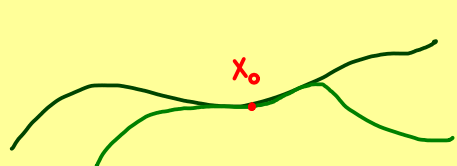
A solution $\alpha: I \rightarrow \mathbb{D}$ is called maximal if there is no extension.

Proposition:

(IVP $_{x_0}^{t_0}$) for $v: \mathbb{D} \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous

has exactly one maximal solution (defined on an open interval).

Proof:



$$\alpha_1: I_1 \rightarrow \mathbb{D}$$

$$\alpha_2: I_2 \rightarrow \mathbb{D}$$

two solutions of (IVP $_{x_0}^{t_0}$)

$$\Rightarrow I := I_1 \cap I_2 = (a, b)$$

$$\Rightarrow \alpha_1|_I, \alpha_2|_I \text{ two solutions of (IVP}_{x_0}^{t_0})$$

There is $\varepsilon > 0$ such that $\alpha_1|_{(t_0 - \varepsilon, t_0 + \varepsilon)} = \alpha_2|_{(t_0 - \varepsilon, t_0 + \varepsilon)}$

$$\left\{ J \text{ open interval} \mid I \supseteq J \supseteq (t_0 - \varepsilon, t_0 + \varepsilon) \text{ with } \alpha_1|_J = \alpha_2|_J \right\} = \mathcal{M}$$

$$(t_-, t_+) := \bigcup_{J \in \mathcal{M}} J \text{ gives maximal open interval}$$

Show: $t_+ = b$

Assume: $t_+ \neq b$

Then: $\alpha_1(t) = \alpha_2(t)$ for all $t \in (t_-, t_+)$

$$\downarrow t \rightarrow t_+ \quad \downarrow$$

$$\tilde{x}_0 = \alpha_1(t_+) = \alpha_2(t_+) \text{ because of continuity on } I$$

Look at (IVP $_{\tilde{x}_0}^{t_+}$): uniqueness of solution implies:

$$\alpha_1(t) = \alpha_2(t) \text{ for } t \in (t_+ - \tilde{\varepsilon}, t_+ + \tilde{\varepsilon})$$

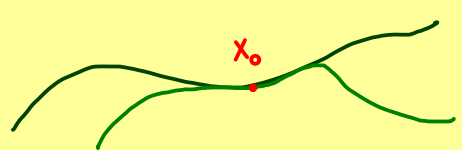
$$\Rightarrow (t_-, t_+ + \tilde{\varepsilon}) \in \mathcal{M} \quad \downarrow$$

Conclusion: $(t_-, t_+) = I$ and

$$\alpha_1|_I = \alpha_2|_I \Rightarrow \alpha: I_1 \cup I_2 \rightarrow \mathbb{D}$$

Define: $\left\{ I \text{ open interval} \mid \text{there is a solution } \alpha: I \rightarrow \mathbb{D} \text{ for (IVP}_{x_0}^{t_0}) \right\} = \mathcal{S}$

$$\bigcup_{I \in \mathcal{S}} I \text{ open interval for maximal solution} \quad \square$$



cannot happen!

Definition:

If the maximal solution is defined on $I = \mathbb{R}$, then it's called a global solution.

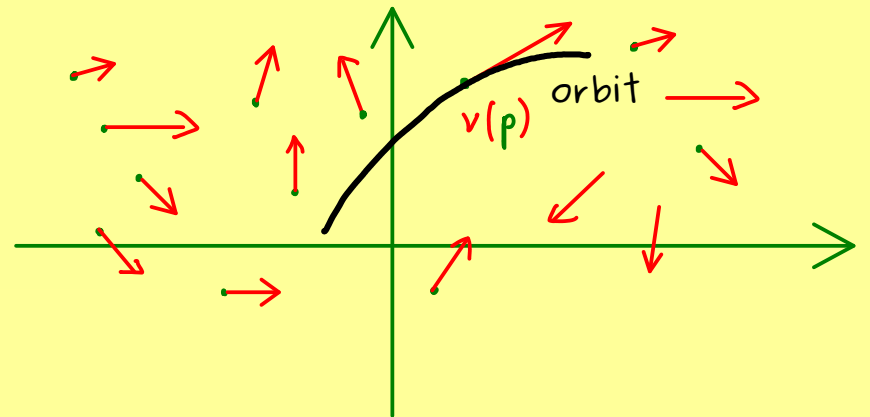


Ordinary Differential Equations – Part 15

$$\dot{x} = v(x) \quad \text{vector field}$$

$$v: \mathcal{D} \rightarrow \mathbb{R}^n$$

open in $\mathbb{R}^n$



For $v: \mathcal{D} \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous:

(IVP $_{x_0}^{t_0}$)

$$\begin{cases} \dot{x} = v(x) \\ x(t_0) = x_0 \end{cases}$$

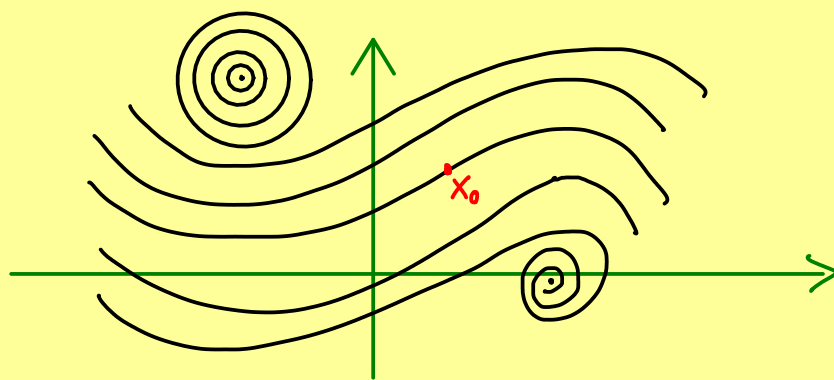
has a unique maximal solution $\alpha: \mathcal{I} \rightarrow \mathcal{D}$



$$\beta(\tilde{t}) := \alpha(\tilde{t} + t_0)$$

$\beta: \tilde{\mathcal{I}} \rightarrow \mathcal{D}$ is a maximal solution (IVP $_{x_0}^0$)

Phase portrait:



orbit at x_0

$$\left\{ \alpha(t) \mid t \in \mathcal{I} \text{ where } \alpha: \mathcal{I} \rightarrow \mathcal{D} \right\}$$

is the max. solution of (IVP $_{x_0}^0$)

Proposition:

For $v: \mathcal{D} \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous, the phase portrait satisfies:

(a) For all $x \in \mathcal{D}$ there is an orbit $\mathcal{O} \ni x$.

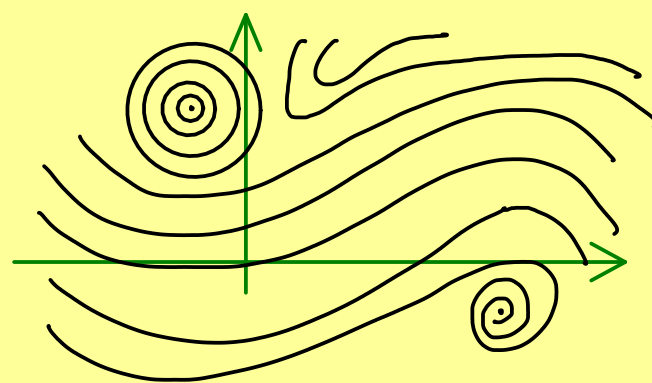
(b) Two orbits $\mathcal{O}_1, \mathcal{O}_2$ satisfy: $\mathcal{O}_1 \cap \mathcal{O}_2 \neq \emptyset \Rightarrow \mathcal{O}_1 = \mathcal{O}_2$



Ordinary Differential Equations – Part 16

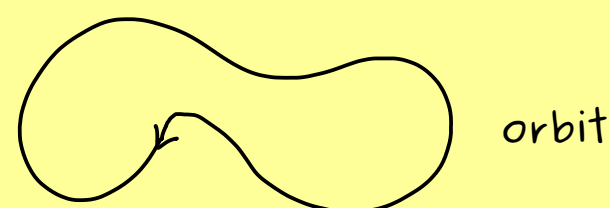
$$\dot{x} = v(x), \quad v: \mathcal{D} \rightarrow \mathbb{R}^n$$

← open in $\mathbb{R}^n$



Definition: A global solution $\alpha: \mathbb{R} \rightarrow \mathcal{D}$ of $\dot{x} = v(x)$ is called:

- fixed point if $\alpha(t) = \alpha(0)$ for all $t \in \mathbb{R}$.
 - periodic if there is a $T > 0$ with $\alpha(t+T) = \alpha(t)$ for all $t \in \mathbb{R}$.
- a period

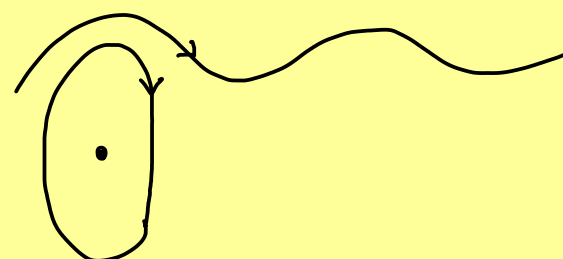


Proposition: For $v: \mathcal{D} \rightarrow \mathbb{R}^n$ loc. Lipschitz continuous,

there are three options for the maximal solution α of

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases}$$

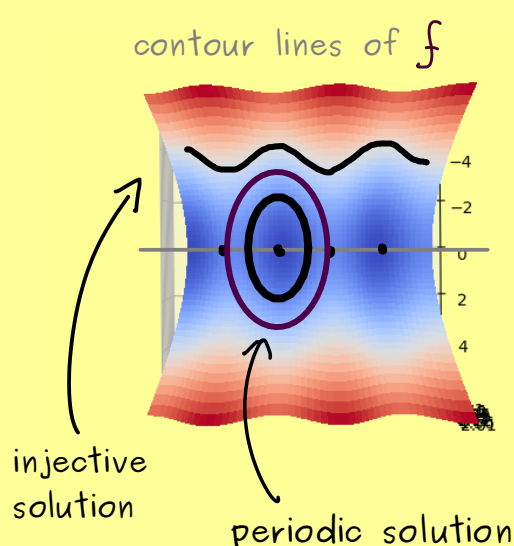
- α is injective
- α is fixed point
- α is periodic



Example:

$$\ddot{x} = -\sin(x) \rightsquigarrow \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} x_2 \\ -\sin(x_1) \end{pmatrix} = v(x_1, x_2)$$

Do we have $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ with $f(\alpha(t)) = \text{constant}$ for all t ?



Note: $f(\alpha(t)) = \text{constant}$ for all t

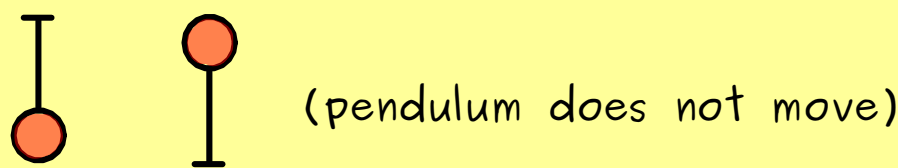
$$\Leftrightarrow \frac{d}{dt} f(\alpha(t)) = 0 \quad \text{for all } t$$

chain rule

$$\Leftrightarrow \langle \text{grad} f(\alpha(t)), \underbrace{\dot{\alpha}(t)}_{v(\alpha(t))} \rangle = 0 \quad \text{for all } t$$

$$f(x_1, x_2) = \frac{1}{2} x_2^2 - \cos(x_1) \quad \text{satisfies} \quad \langle \text{grad} f(x_1, x_2), v(x_1, x_2) \rangle = 0.$$

Fixed point: $\text{grad} f(x_1, x_2) = \begin{pmatrix} \sin(x_1) \\ x_2 \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Leftrightarrow x_2 = 0, \quad x_1 = k \cdot \pi$
 $k \in \mathbb{Z}$





Ordinary Differential Equations - Part 17

In part 12: Picard-Lindelöf theorem for initial value problem

$$\begin{cases} \dot{x} = v(x) \\ x(0) = x_0 \end{cases} \quad \begin{array}{l} \text{locally Lipschitz continuous} \\ \Rightarrow \text{there is a unique solution} \end{array}$$

Now:

$$\begin{cases} \dot{x} = w(t, x) \\ x(t_0) = x_0 \end{cases} \quad \begin{array}{l} \text{initial value problem} \\ \text{continuous function } w: \mathcal{I} \times \mathcal{U} \rightarrow \mathbb{R}^n \end{array}$$

interval in $\mathbb{R}$ open set in $\mathbb{R}^n$

generalized rectangle in $\mathbb{R}^{n+1}$

Picard-Lindelöf theorem (for non-autonomous systems)

Assume $w: \mathcal{I} \times \mathcal{U} \rightarrow \mathbb{R}^n$ satisfies: $\forall \mathcal{K} \subseteq \mathcal{I} \times \mathcal{U}$ compact $\exists L_{\mathcal{K}} > 0 \forall (t, x), (t, y) \in \mathcal{K}$:

continuous!

$$\|w(t, x) - w(t, y)\| \leq L_{\mathcal{K}} \cdot \|x - y\|$$

standard norm in $\mathbb{R}^n$

Then: For $x_0 \in \mathcal{U}$, there is $\epsilon > 0$ and a unique solution $\alpha: (t_0 - \epsilon, t_0 + \epsilon) \rightarrow \mathcal{U}$

for the initial value problem

$$\begin{cases} \dot{x} = w(t, x) \\ x(t_0) = x_0 \end{cases}.$$

Proof: Same as in part 12 with

$$\Phi(\alpha)(t) = x_0 + \int_{t_0}^t w(s, \alpha(s)) ds$$

and Banach fixed-point theorem.

Picard-Lindelöf theorem (special version)

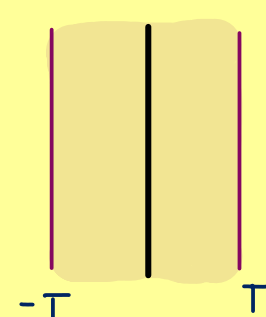
Assume $w: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous and satisfies: for each $T > 0$:

$$\exists L_T > 0 \forall t \in [-T, T] \forall x, y \in \mathbb{R}^n: \|w(t, x) - w(t, y)\| \leq L_T \cdot \|x - y\|$$

Then there is a unique global solution $\alpha: \mathbb{R} \rightarrow \mathbb{R}^n$

for the initial value problem

$$\begin{cases} \dot{x} = w(t, x) \\ x(t_0) = x_0 \end{cases}.$$



Proof: Set $t_0 = 0$. Complete metric space $X = C([-T, T], \mathbb{R}^n)$

with metric $d(\alpha, \beta) := \sup_{t \in [-T, T]} e^{-2L_T |t|} \cdot \|\alpha(t) - \beta(t)\|_{\mathbb{R}^n}$

$$\Phi(\alpha)(t) = x_0 + \int_0^t w(s, \alpha(s)) ds$$

standard norm

$$\begin{aligned} d(\Phi(\alpha), \Phi(\beta)) &= \sup_{t \in [-T, T]} e^{-2L_T |t|} \|\Phi(\alpha)(t) - \Phi(\beta)(t)\|_{\mathbb{R}^n} \\ &= \sup_{t \in [-T, T]} e^{-2L_T |t|} \left\| \int_0^t (w(s, \alpha(s)) - w(s, \beta(s))) ds \right\|_{\mathbb{R}^n} \\ &\stackrel{\text{triangle inequality for integrals}}{\leq} \sup_{t \in [-T, T]} e^{-2L_T |t|} \left| \int_0^t \|w(s, \alpha(s)) - w(s, \beta(s))\|_{\mathbb{R}^n} ds \right| \\ &\leq \sup_{t \in [-T, T]} e^{-2L_T |t|} \left| \int_0^t L_T \cdot e^{2L_T |s|} e^{-2L_T |s|} \|\alpha(s) - \beta(s)\|_{\mathbb{R}^n} ds \right| \\ &\leq \sup_{t \in [-T, T]} e^{-2L_T |t|} L_T \cdot d(\alpha, \beta) \left| \int_0^t e^{2L_T |s|} ds \right| \leq d(\alpha, \beta) \\ &\leq \frac{1}{2} d(\alpha, \beta) \underbrace{\sup_{t \in [-T, T]} (1 - e^{-2L_T |t|})}_{\leq 1} \frac{1}{2L_T} (e^{2L_T |t|} - 1) \\ &\leq \frac{1}{2} d(\alpha, \beta) \end{aligned}$$

Banach fixed-point theorem:

$\Rightarrow \Phi: X \rightarrow X$ is a contraction $\Rightarrow$ unique solution $\alpha: [-T, T] \rightarrow \mathbb{R}^n$
for all $T > 0$

$\Rightarrow$ global solution $\alpha: \mathbb{R} \rightarrow \mathbb{R}^n$

□



Ordinary Differential Equations - Part 18

Definition: A system of ODEs $\dot{x} = w(t, x)$

is called a system of linear differential equations if

$$W : \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n, \quad (t, x) \longmapsto A(t)x + b(t)$$

$\swarrow$ continuous $\nwarrow$ or interval $\swarrow$ or open set

with $A : t \mapsto A(t) \in \mathbb{R}^{n \times n}$ $b : t \mapsto b(t) \in \mathbb{R}^n$
 $\nwarrow$ continuous $\swarrow$ continuous

Note: • If $b(t) = 0$ for all t , then the system is called homogeneous.

• If $A(t) = A$, $b(t) = b$ for all t , then the system is called autonomous.

Lipschitz condition?

$$\begin{aligned} \|w(t, x) - w(t, y)\| &= \|A(t)x + b(t) - (A(t)y + b(t))\| \\ &= \|A(t)(x - y)\| \leq \|A(t)\| \cdot \|x - y\| \\ &\leq L_T \cdot \|x - y\| \end{aligned}$$

$\nwarrow$ matrix norm/ operator norm
 $[-T, T] \ni t \mapsto \|A(t)\|$ continuous

Picard-Lindelöf theorem
(special version)

$\implies$ unique global solution $\alpha : \mathbb{R} \longrightarrow \mathbb{R}^n$ for initial value problem

$$\begin{cases} \dot{x} = w(t, x) \\ x(t_0) = x_0 \end{cases}.$$

Example:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2t \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ e^{t^2} \end{pmatrix}$$

$$\leadsto \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} - \begin{pmatrix} 1 \cdot x_1 \\ 2t \cdot x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{t^2} \end{pmatrix} \begin{matrix} \leftarrow \bar{e}^{-t} \\ \leftarrow \bar{e}^{-t^2} \end{matrix}$$

$$\leadsto \begin{cases} \dot{x}_1 \bar{e}^{-t} - x_1 \bar{e}^{-t} = 0 \\ \dot{x}_2 \bar{e}^{-t^2} - x_2 \cdot 2t \bar{e}^{-t^2} = 1 \end{cases} \leadsto \begin{cases} \frac{d}{dt}(x_1 \bar{e}^{-t}) = 0 \\ \frac{d}{dt}(x_2 \bar{e}^{-t^2}) = 1 \end{cases}$$

$$\begin{aligned} \Rightarrow \begin{cases} x_1 \bar{e}^{-t} = c_1 \\ x_2 \bar{e}^{-t^2} = t + c_2 \end{cases} &\Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}(t) = \begin{pmatrix} c_1 e^t \\ (t + c_2) e^{t^2} \end{pmatrix} \\ &= \begin{pmatrix} e^t & 0 \\ 0 & e^{t^2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ t e^{t^2} \end{pmatrix} \end{aligned}$$



Ordinary Differential Equations - Part 19

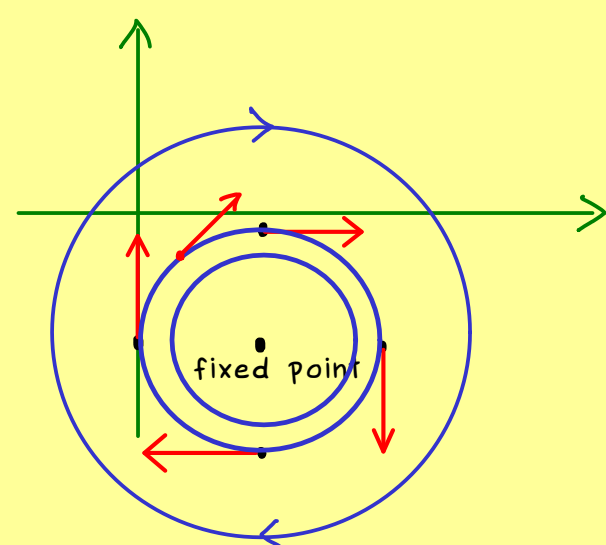
System of linear differential equations: (of first order)

$$\dot{x} = A(t)x + b(t) \quad \text{with} \quad \begin{array}{l} \text{interval} \\ \text{in } \mathbb{R} \end{array} \begin{array}{l} I \ni t \longrightarrow A(t) \in \mathbb{R}^{n \times n} \\ I \ni t \longrightarrow b(t) \in \mathbb{R}^n \end{array} \quad \begin{array}{l} \text{continuous} \end{array}$$

- solutions are global $\alpha: I \rightarrow \mathbb{R}^n$
- autonomous systems: $A(t) = A$, $b(t) = b$

example: $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$v(x) = Ax + b$$



- corresponding homogeneous system

$$\dot{x} = A(t)x$$

Fact: If $\alpha: I \rightarrow \mathbb{R}^n$, $\beta: I \rightarrow \mathbb{R}^n$ are two solutions of $\dot{x} = A(t)x$,

$$(\alpha + \beta)'(t) = \dot{\alpha}(t) + \dot{\beta}(t) = A(t)\alpha(t) + A(t)\beta(t)$$

$$= A(t)(\alpha(t) + \beta(t))$$

$$(\lambda \cdot \alpha)'(t) = A(t)(\lambda \cdot \alpha(t)) \quad \leadsto$$

linear combinations of solutions are solutions

Proposition: The solution set of the corresponding homogeneous system

$$S_0 := \left\{ \alpha: I \rightarrow \mathbb{R}^n \text{ continuously differentiable} \mid \dot{\alpha}(t) = A(t)\alpha(t) \text{ for all } t \in I \right\}$$

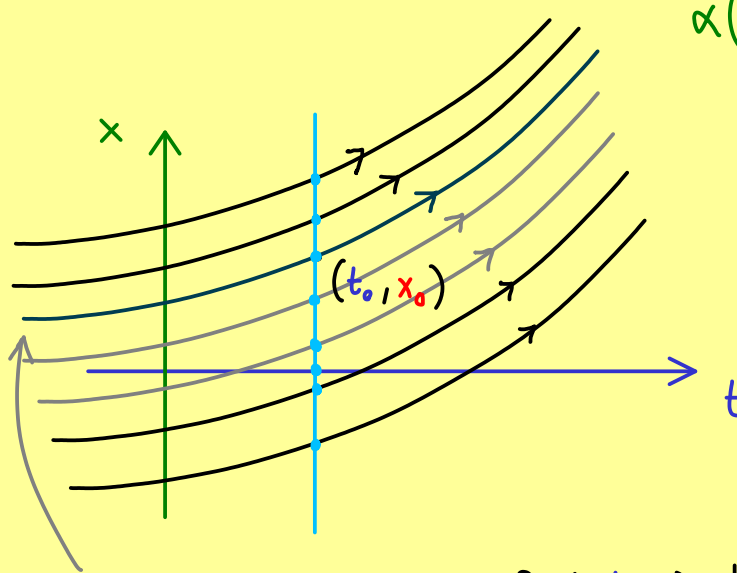
forms an n -dimensional $\mathbb{R}$ -vector space.

Proof: S_0 is a subspace in the $\mathbb{R}$ -vector space $C^1(I, \mathbb{R}^n)$.

What about the dimension of S_0 ?

(IVP $\begin{smallmatrix} t_0 \\ x_0 \end{smallmatrix}$) $\begin{array}{l} \dot{x} = A(t)x \\ x(t_0) = x_0 \end{array}$ $\xrightarrow{\text{Picard-Lindelöf theorem (special version)}}$ unique solution $\alpha: I \rightarrow \mathbb{R}^n$ $\alpha(t_0) = x_0$

extended phase portrait:



extended orbit at $(t_0, x_0) : \left\{ \begin{pmatrix} t \\ \alpha(t) \end{pmatrix} \mid t \in I \text{ where } \alpha \text{ is unique solution of (IVP } \begin{smallmatrix} t_0 \\ x_0 \end{smallmatrix}) \right\}$

define a map: $\ell: S_0 \rightarrow \mathbb{R}^n$
 $\alpha \mapsto \alpha(t_0)$ $\longleftarrow$ linear map!

$\hookrightarrow$ surjective (every (IVP $\begin{smallmatrix} t_0 \\ x_0 \end{smallmatrix}$) has a solution)

$\hookrightarrow$ injective $\left(\ell(\alpha) = \ell(\beta) \Rightarrow \alpha(t_0) = \beta(t_0) \right)$
 $\xRightarrow{\text{uniqueness}} \alpha = \beta \text{ on } I$

$\Rightarrow \ell: S_0 \rightarrow \mathbb{R}^n$ isomorphism

$\Rightarrow \dim(S_0) = \dim(\mathbb{R}^n) = n$

□



Ordinary Differential Equations - Part 20

System of linear differential equations (homogeneous + autonomous)

$$(\text{IVP}_{x_0}^0) \quad \begin{cases} \dot{x} = A x \\ x(0) = x_0 \end{cases}, \quad \begin{matrix} A \in \mathbb{R}^{n \times n} \\ x_0 \in \mathbb{R}^n \end{matrix}$$

↳ Picard iteration (see part 13)

start with a guess $\tilde{\alpha}: \mathbb{R} \rightarrow \mathbb{R}^n$

$$\Phi(\tilde{\alpha})(t) = x_0 + \int_0^t A \tilde{\alpha}(s) ds \rightsquigarrow \Phi^n(\tilde{\alpha}) \xrightarrow{n \rightarrow \infty} \alpha \quad \uparrow \text{solution of } (\text{IVP}_{x_0}^0)$$

Picard iteration: guess: $\tilde{\alpha}: \mathbb{R} \rightarrow \mathbb{R}^n$, $\tilde{\alpha}(t) = x_0$

$$1^{\text{st}} \text{ step: } \Phi(\tilde{\alpha})(t) = x_0 + \int_0^t A x_0 ds = (\mathbb{1} + tA) x_0$$

$$2^{\text{nd}} \text{ step: } \Phi^2(\tilde{\alpha})(t) = x_0 + \int_0^t A (\mathbb{1} + sA) x_0 ds \\ = x_0 + tAx_0 + \frac{1}{2}t^2A^2x_0 = (\mathbb{1} + tA + \frac{1}{2}t^2A^2)x_0$$

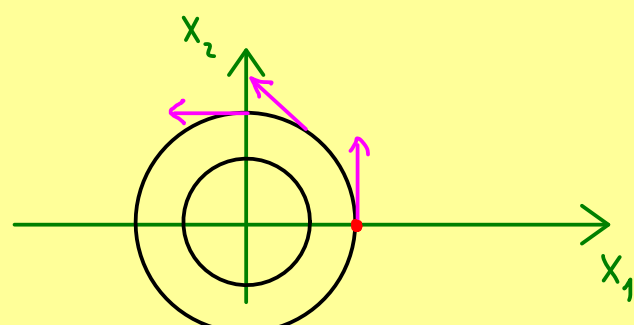
$$n^{\text{th}} \text{ step: } \Phi^n(\tilde{\alpha})(t) = (\mathbb{1} + tA + \frac{1}{2}t^2A^2 + \frac{1}{6}t^3A^3 + \dots + \frac{1}{n!}t^nA^n)x_0$$

$$\xrightarrow{n \rightarrow \infty} \text{solution of } (\text{IVP}_{x_0}^0) \quad \alpha(t) = \sum_{k=0}^{\infty} \frac{(t \cdot A)^k}{k!} x_0 \\ \underbrace{\qquad\qquad\qquad}_{=: \exp(t \cdot A)} = e^{tA}$$

matrix exponential

Example:

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



$$\exp(t \cdot A) = \left(\mathbb{1} + tA + \frac{1}{2}t^2A^2 + \frac{1}{6}t^3A^3 + \frac{1}{4!}t^4A^4 + \dots \right) \\ \underbrace{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}_{=: \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}} \quad \underbrace{\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}}_{=: \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}} \quad \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{=: \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}} \quad \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{=: \mathbb{1}}$$

$$= \begin{pmatrix} \underbrace{1 - \frac{1}{2}t^2 + \frac{1}{4!}t^4 \pm \dots}_{=: \cos(t)} & -\sin(t) \\ \underbrace{0 + t - \frac{1}{6}t^3 + \frac{1}{5!}t^5 \pm \dots}_{=: \sin(t)} & \cos(t) \end{pmatrix}$$

solution of $(\text{IVP}_{x_0}^0)$ with $x_0 = \begin{pmatrix} c \\ 0 \end{pmatrix}$:

$$\alpha(t) = \exp(t \cdot A) x_0 = \begin{pmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{pmatrix} \begin{pmatrix} c \\ 0 \end{pmatrix} = c \cdot \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix}$$

$\Rightarrow$ all orbits are circles!



Ordinary Differential Equations - Part 21

System of linear differential equations:

$$\dot{x} = A(t)x + b(t) \quad (*)$$

with
interval
in $\mathbb{R}$

$$I \ni t$$

continuous

$$A(t) \in \mathbb{R}^{n \times n}$$

$$b(t) \in \mathbb{R}^n$$

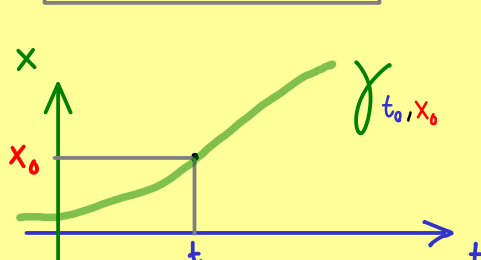
We already know:

- the homogeneous part of $(*)$ $(\dot{x} = A(t)x)$

has an n -dimensional solution space S_0

- the initial value problem (IVP) $\begin{cases} \dot{x} = A(t)x + b(t) \\ x(t_0) = x_0 \end{cases}$

$$\gamma_{t_0, x_0} : I \rightarrow \mathbb{R}^n$$



Solution set:

$$S := \{ \beta : I \rightarrow \mathbb{R}^n \text{ continuously differentiable} \mid \beta \text{ solution of } (*) \}$$

$$S_0 + \gamma_{t_0, x_0} := \{ \alpha + \gamma_{t_0, x_0} \mid \alpha \in S_0 \} \quad (\text{affine subspace})$$

Show $S = S_0 + \gamma_{t_0, x_0} : (\Rightarrow)$ Take $\alpha \in S_0$:

$$\begin{aligned} A(t)(\alpha(t) + \gamma_{t_0, x_0}(t)) + b(t) &= \underbrace{A(t)\alpha(t)}_{\dot{\alpha}(t)} + \underbrace{A(t)\gamma_{t_0, x_0}(t) + b(t)}_{\dot{\gamma}_{t_0, x_0}(t)} \\ &= \dot{\alpha}(t) + \dot{\gamma}_{t_0, x_0}(t) \\ &= (\alpha + \gamma_{t_0, x_0})'(t) \end{aligned}$$

$$\Rightarrow \alpha + \gamma_{t_0, x_0} \in S$$

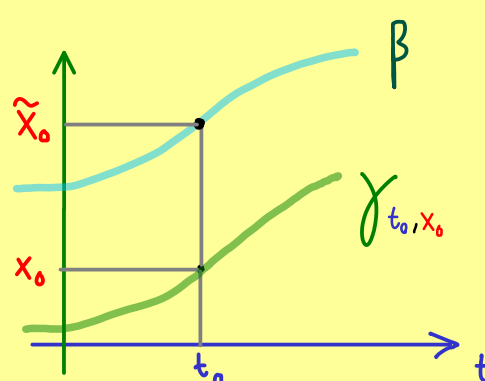
$(\Leftarrow)$ Take $\beta \in S$ and set $\tilde{x}_0 := \beta(t_0)$

$\Rightarrow \beta$ is solution of $(\text{IVP}_{\tilde{x}_0}^{t_0})$

Choose $\alpha \in S_0$ as the solution

of the initial value problem

$$\begin{cases} \dot{x} = A(t)x \\ x(t_0) = \tilde{x}_0 - x_0 \end{cases}$$



Then: $\alpha + \gamma_{t_0, x_0} \in S$ with $(\alpha + \gamma_{t_0, x_0})(t_0) = \alpha(t_0) + \gamma_{t_0, x_0}(t_0)$

$$= \tilde{x}_0 - x_0 + x_0 = \tilde{x}_0$$

$$\Rightarrow \alpha + \gamma_{t_0, x_0} \text{ is solution of } (\text{IVP}_{\tilde{x}_0}^{t_0})$$

uniqueness

$$\Rightarrow \beta = \alpha + \gamma_{t_0, x_0}$$

□

Result:

The solution set of $\dot{x} = A(t)x + b(t)$ is given by

$$S = S_0 + \gamma$$

where S_0 is the solution space of the homogeneous part $\dot{x} = A(t)x$

and γ is a particular solution of $\dot{x} = A(t)x + b(t)$.

$(S \text{ is an } n\text{-dimensional affine subspace})$



Ordinary Differential Equations - Part 22

$$A \in \mathbb{R}^{n \times n} \Rightarrow e^{tA} \in \mathbb{R}^{n \times n} \text{ columns span solution space of } \dot{x} = Ax$$

Remember:

$$\begin{cases} \dot{x} = Ax \\ x(0) = x_0 \end{cases}$$

has unique solution: $t \mapsto e^{tA} x_0$

Definition:

$$e^{tA} := \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k \text{ exists for every } t \in \mathbb{R}, A \in \mathbb{R}^{n \times n}$$

- each component is a function $t \in [a, b] \rightarrow \mathbb{R}$
- we have uniform convergence

Properties: (a) derivative of the matrix exponential:

$$\begin{aligned} \frac{d}{dt} e^{tA} &:= \lim_{h \rightarrow 0} \frac{e^{(t+h)A} - e^{tA}}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\sum_{k=0}^{\infty} \frac{(t+h)^k}{k!} A^k - \sum_{k=0}^{\infty} \frac{t^k}{k!} A^k \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} \left((t+h)^k - t^k \right) \right) \\ &\stackrel{\text{uniform convergence}}{=} \sum_{k=0}^{\infty} \frac{A^k}{k!} \underbrace{\lim_{h \rightarrow 0} \frac{(t+h)^k - t^k}{h}}_{= \begin{cases} k \cdot t^{k-1}, & k \geq 1 \\ 0, & k = 0 \end{cases}} \\ &= \sum_{k=1}^{\infty} \frac{A^k}{k!} \cdot k \cdot t^{k-1} \\ &= \sum_{k=1}^{\infty} \frac{t^{k-1}}{(k-1)!} A^{k-1} \cdot A = e^{tA} A = A e^{tA} \end{aligned}$$

(b) exponentiation identity: $e^{A+B} = e^A e^B$ for matrices with $AB = BA$

$$(c) \text{ inverse: } \left. \begin{aligned} e^A e^{-A} &= e^{A-A} = e^0 = \mathbb{1} \\ e^{-A} e^A &= e^{A-A} = e^0 = \mathbb{1} \end{aligned} \right\} (e^A)^{-1} = e^{-A}$$



Ordinary Differential Equations - Part 23

Example: System of linear differential equations (homogeneous + autonomous)

$$\begin{aligned} \dot{x}_1 &= -x_1 + 3x_2 \\ \dot{x}_2 &= x_1 + x_2 \end{aligned} \quad \Leftrightarrow \quad \dot{\mathbf{x}} = \underbrace{\begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix}}_{\mathbf{A}} \mathbf{x}$$

General solution: $e^{t\mathbf{A}} = \sum_{k=0}^{\infty} \frac{(t\mathbf{A})^k}{k!}$
(columns span solution space)

Remark: If $\mathbf{B} = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$, then $\mathbf{B}^k = \begin{pmatrix} \lambda^k & 0 \\ 0 & \mu^k \end{pmatrix}$ and $e^{\mathbf{B}} = \begin{pmatrix} e^\lambda & 0 \\ 0 & e^\mu \end{pmatrix}$

If $\mathbf{A}$ is diagonalizable, then $\mathbf{A} = \mathbf{X} \mathbf{D} \mathbf{X}^{-1}$, $\mathbf{A}^2 = \mathbf{X} \mathbf{D} \mathbf{X}^{-1} \mathbf{X} \mathbf{D} \mathbf{X}^{-1} = \mathbf{X} \mathbf{D}^2 \mathbf{X}^{-1}$
 $\mathbf{A}^k = \mathbf{X} \mathbf{D}^k \mathbf{X}^{-1}$

$$\Rightarrow e^{t\mathbf{A}} = \mathbf{X} \cdot e^{t\mathbf{D}} \mathbf{X}^{-1}$$

Back to the example:

$$\mathbf{A} = \begin{pmatrix} -1 & 3 \\ 1 & 1 \end{pmatrix}$$

eigenvalues: $0 = \det(\mathbf{A} - \lambda \cdot \mathbf{1}) = \det \begin{pmatrix} -1-\lambda & 3 \\ 1 & 1-\lambda \end{pmatrix} = \lambda^2 - 4$
 $\Rightarrow \lambda_1 = -2, \lambda_2 = 2$

eigenvectors: $\text{Ker}(\mathbf{A} - \lambda_1 \cdot \mathbf{1}) = \text{Ker} \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix} = \text{Span} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$

$$\text{Ker}(\mathbf{A} - \lambda_2 \cdot \mathbf{1}) = \text{Ker} \begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix} = \text{Span} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

form invertible matrix: $\mathbf{X} = \begin{pmatrix} -3 & 1 \\ 1 & 1 \end{pmatrix} \leadsto \mathbf{X}^{-1} = \frac{1}{4} \begin{pmatrix} -1 & 1 \\ 1 & 3 \end{pmatrix}$

diagonalization: $\mathbf{A} = \mathbf{X} \mathbf{D} \mathbf{X}^{-1}$ with $\mathbf{D} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}$

matrix exponential: $e^{t\mathbf{A}} = \mathbf{X} \cdot e^{t\mathbf{D}} \mathbf{X}^{-1} = \mathbf{X} \begin{pmatrix} e^{-2t} & 0 \\ 0 & e^{2t} \end{pmatrix} \mathbf{X}^{-1}$
$$= \frac{1}{4} \begin{pmatrix} 3e^{-2t} + e^{2t} & -3e^{-2t} + 3e^{2t} \\ -e^{-2t} + e^{2t} & e^{-2t} + 3e^{2t} \end{pmatrix}$$

solution of initial value problem:

$$\begin{aligned} \begin{cases} \dot{\mathbf{x}} = \mathbf{A} \mathbf{x} \\ \mathbf{x}(0) = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \end{cases} &\leadsto \boldsymbol{\alpha}(t) = e^{t\mathbf{A}} \begin{pmatrix} 0 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} -3e^{-2t} + 3e^{2t} \\ e^{-2t} + 3e^{2t} \end{pmatrix} \end{aligned}$$



Ordinary Differential Equations - Part 24

System of linear ODEs (homogeneous + autonomous)

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} \quad \leadsto \quad \text{solutions } t \mapsto e^{t\mathbf{A}} \cdot \mathbf{x}_0 \quad \text{for } \mathbf{x}_0 \in \mathbb{R}^n$$

easy to calculate if $\mathbf{A}$ has n different eigenvalues:
 $\lambda_1, \lambda_2, \dots, \lambda_n$

Linear ODE of order n (homogeneous + autonomous)

$$x^{(n)} + a_{n-1} x^{(n-1)} + a_{n-2} x^{(n-2)} + \dots + a_1 \dot{x} + a_0 x = 0$$

Transform into system of first order:

$$\begin{pmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \vdots \\ \dot{y}_n \end{pmatrix} = \begin{pmatrix} x \\ \dot{x} \\ \vdots \\ x^{(n-1)} \end{pmatrix} \quad \text{and} \quad \dot{y}_n = -a_{n-1} y_n - \dots - a_1 y_2 - a_0 y_1$$

$$\Rightarrow \quad \dot{\mathbf{y}} = \begin{pmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \vdots \\ \dot{y}_n \end{pmatrix} = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \\ \dot{y}_n \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 1 \\ -a_0 & -a_1 & \dots & -a_{n-1} \end{pmatrix}}_{\mathbf{A} \text{ (n \times n)-matrix}} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

General solution: $e^{t\mathbf{A}} \iff$ eigenvalues of $\mathbf{A}$

Characteristic polynomial: $\det(\mathbf{A} - \lambda \cdot \mathbb{1}) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + a_{n-2} \lambda^{n-2} + \dots + a_1 \lambda + a_0)$

is called the characteristic polynomial of the ODE

$$x^{(n)} + a_{n-1} x^{(n-1)} + a_{n-2} x^{(n-2)} + \dots + a_1 \dot{x} + a_0 x = 0$$

Rule of thumb: Use approach $x(t) = e^{\lambda t}$

Result: If the characteristic polynomial has n different zeros in the real numbers

$\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$, then:

$$e^{\lambda_1 t} \cdot \begin{pmatrix} 1 \\ \lambda_1 \\ \lambda_1^2 \\ \vdots \\ \lambda_1^{n-1} \end{pmatrix}, e^{\lambda_2 t} \cdot \begin{pmatrix} 1 \\ \lambda_2 \\ \lambda_2^2 \\ \vdots \\ \lambda_2^{n-1} \end{pmatrix}, \dots, e^{\lambda_n t} \cdot \begin{pmatrix} 1 \\ \lambda_n \\ \lambda_n^2 \\ \vdots \\ \lambda_n^{n-1} \end{pmatrix} \quad \text{span solution space of}$$

$$\dot{\mathbf{y}} = \mathbf{A} \mathbf{y}$$

and $t \mapsto e^{\lambda_1 t}, t \mapsto e^{\lambda_2 t}, \dots, t \mapsto e^{\lambda_n t}$ span solution space of

$$x^{(n)} + a_{n-1} x^{(n-1)} + a_{n-2} x^{(n-2)} + \dots + a_1 \dot{x} + a_0 x = 0$$



Ordinary Differential Equations - Part 25

We know: $\dot{x} = A x$, $x(0) = x_0 \in \mathbb{R}^n \Rightarrow$ unique solution $t \mapsto e^{tA} x_0$

A is $(n \times n)$ -matrix

How to calculate if A is not diagonalizable?

Jordan normal form!

Example: $\dot{x} = A x$ with

$$A = \begin{pmatrix} \boxed{\begin{matrix} 2 & & \\ & 2 & 1 \\ & & 2 \end{matrix}} & \begin{matrix} \\ \\ \end{matrix} \\ \begin{matrix} 3 & 1 \\ & 3 & 1 \\ & & 3 \end{matrix} & \begin{matrix} \\ \\ \end{matrix} \end{pmatrix}$$

Jordan block 1 contains two Jordan boxes

Jordan block 2 contains one Jordan box

Powers of A :

$$A^k = \begin{pmatrix} \boxed{\begin{matrix} 2 & & \\ & 2 & 1 \\ & & 2 \end{matrix}}^k & \begin{matrix} \\ \\ \end{matrix} \\ \begin{matrix} 3 & 1 \\ & 3 & 1 \\ & & 3 \end{matrix}^k & \begin{matrix} \\ \\ \end{matrix} \end{pmatrix}$$

Jordan block 2: $\begin{pmatrix} 3 & 1 \\ & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ & 3 \end{pmatrix} = \begin{pmatrix} 3^2 & 3+3 & 1 \\ & 3^2 & 3+3 \\ & & 3^2 \end{pmatrix} \leadsto$ triangular form remains

Let's split it up: $\begin{pmatrix} 3 & 1 \\ & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 3 & \\ & 3 \end{pmatrix}}_{\mathcal{D} \text{ (diagonal)}} + \underbrace{\begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix}}_{\mathcal{N} \text{ (nilpotent)}}$

$\leadsto \mathcal{N}^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\mathcal{N}^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\mathcal{D}\mathcal{N} = \begin{pmatrix} 3 & \\ & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

$\mathcal{N}\mathcal{D} = \begin{pmatrix} 0 & 1 \\ & 0 \end{pmatrix} \begin{pmatrix} 3 & \\ & 3 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

they commute!

$\Rightarrow e^{t(\mathcal{D}+\mathcal{N})} = e^{t\mathcal{D}} \cdot e^{t\mathcal{N}}$

easy because $\mathcal{D}$ is diagonal

easy because $\mathcal{N}$ is nilpotent

$= \begin{pmatrix} e^{3t} & & \\ & e^{3t} & \\ & & e^{3t} \end{pmatrix} \cdot \left(\mathbb{1} + t\mathcal{N} + \frac{1}{2}t^2\mathcal{N}^2 \right) = \begin{pmatrix} e^{3t} & & \\ & e^{3t} & \\ & & e^{3t} \end{pmatrix} \cdot \begin{pmatrix} 1 & t & \frac{1}{2}t^2 \\ & 1 & t \\ & & 1 \end{pmatrix}$

$= \begin{pmatrix} e^{3t} & t e^{3t} & \frac{1}{2}t^2 e^{3t} \\ 0 & e^{3t} & t e^{3t} \\ 0 & 0 & e^{3t} \end{pmatrix}$

Jordan block 1 $\begin{pmatrix} 2 & & \\ & 2 & 1 \\ & & 2 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & & \\ & 2 & \\ & & 2 \end{pmatrix}}_{\tilde{\mathcal{D}}} + \underbrace{\begin{pmatrix} 0 & 1 \\ & 0 & 1 \\ & & 0 \end{pmatrix}}_{\tilde{\mathcal{N}}} \leadsto \tilde{\mathcal{N}}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\Rightarrow e^{t(\tilde{\mathcal{D}}+\tilde{\mathcal{N}})} = \begin{pmatrix} e^{2t} & & \\ & e^{2t} & \\ & & e^{2t} \end{pmatrix} (\mathbb{1} + t\tilde{\mathcal{N}})$

$= \begin{pmatrix} e^{2t} & & \\ & e^{2t} & t e^{2t} \\ & & e^{2t} \end{pmatrix}$

Result: $e^{tA} = \begin{pmatrix} e^{2t} & & & & \\ & e^{2t} & t e^{2t} & & \\ & & e^{2t} & & \\ & & & e^{3t} & t e^{3t} \\ & & & & e^{3t} & \frac{1}{2}t^2 e^{3t} \\ & & & & & e^{3t} \end{pmatrix}$



Ordinary Differential Equations - Part 26

System of linear ODEs: $\dot{x} = \underbrace{A}_{\in \mathbb{R}^{n \times n}} x \implies$ Solution space $\mathcal{S}_0$ is an n -dimensional real vector space and the columns of e^{tA} span it.

Remember: If $C \in \mathbb{R}^{n \times n}$ is invertible, then $e^{tA}C$ also spans $\mathcal{S}_0$.

Two-dimensional cases: $A \in \mathbb{R}^{2 \times 2}$

- (1) Two different eigenvalues $\lambda_1 \neq \lambda_2$ $\begin{cases} (1a) \lambda_1, \lambda_2 \in \mathbb{R} \\ (1b) \lambda_1, \lambda_2 \in \mathbb{C} \setminus \mathbb{R} \end{cases}$
- (2) Only one eigenvalue $\lambda_1 = \lambda_2 =: \lambda$ $\begin{cases} (2a) A \text{ is diagonalizable} \\ (2b) A \text{ is not diagonalizable} \end{cases}$

(1a) $\lambda_1 \neq \lambda_2 \in \mathbb{R}$

There is a basis of eigenvectors $(\underbrace{u^{(1)}}_{\in \mathbb{R}^2}, \underbrace{u^{(2)}}_{\in \mathbb{R}^2}) \rightsquigarrow U = \begin{pmatrix} | & | \\ u^{(1)} & u^{(2)} \\ | & | \end{pmatrix} \in \mathbb{R}^{2 \times 2}$ invertible

$$U^{-1}AU = \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix} \rightsquigarrow A = U \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix} U^{-1}$$

$$\rightsquigarrow e^{tA} = U \begin{pmatrix} e^{t\lambda_1} & \\ & e^{t\lambda_2} \end{pmatrix} U^{-1}$$

$$\rightsquigarrow e^{tA}U = U \begin{pmatrix} e^{t\lambda_1} & \\ & e^{t\lambda_2} \end{pmatrix} = \underbrace{\begin{pmatrix} e^{t\lambda_1}u^{(1)} & e^{t\lambda_2}u^{(2)} \\ | & | \end{pmatrix}}_{\text{columns span } \mathcal{S}_0}$$

(1b) $\lambda_1 \neq \lambda_2 \in \mathbb{C} \setminus \mathbb{R}$

$$A \in \mathbb{R}^{2 \times 2} \implies \begin{cases} \lambda_1 = \alpha + i\beta \\ \lambda_2 = \alpha - i\beta \end{cases} \quad (\alpha, \beta \in \mathbb{R}) \quad \begin{matrix} u^{(1)} = u + i v \\ u^{(2)} = u - i v \end{matrix} \quad \begin{matrix} \\ \text{eigenvectors} \end{matrix}$$

"diagonalization" just with real numbers: $\tilde{U}^{-1}A\tilde{U} = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$ with $\tilde{U} = \begin{pmatrix} | & | \\ u & v \\ | & | \end{pmatrix}$

redo for e^{tA} : eigenvalues $e^{t\lambda_1} = e^{t\alpha}(\cos(\beta t) + i \sin(\beta t))$
 $e^{t\lambda_2} = e^{t\alpha}(\cos(\beta t) - i \sin(\beta t))$

$$\implies \tilde{U}^{-1}e^{tA}\tilde{U} = \begin{pmatrix} e^{t\alpha} \cos(\beta t) & e^{t\alpha} \sin(\beta t) \\ -e^{t\alpha} \sin(\beta t) & e^{t\alpha} \cos(\beta t) \end{pmatrix}$$

$$\implies e^{tA}\tilde{U} = e^{t\alpha}\tilde{U} \begin{pmatrix} \cos(\beta t) & \sin(\beta t) \\ -\sin(\beta t) & \cos(\beta t) \end{pmatrix}$$

$$= e^{t\alpha} \underbrace{\begin{pmatrix} \cos(\beta t)u - \sin(\beta t)v & \cos(\beta t)v + \sin(\beta t)u \\ | & | \end{pmatrix}}_{\text{columns span } \mathcal{S}_0}$$

(2a) $\lambda_1 = \lambda_2 =: \lambda$ (A is diagonalizable)

There is a basis of eigenvectors $(u^{(1)}, u^{(2)}) \rightsquigarrow U = \begin{pmatrix} | & | \\ u^{(1)} & u^{(2)} \\ | & | \end{pmatrix}$ invertible

$$U^{-1}AU = \begin{pmatrix} \lambda & \\ & \lambda \end{pmatrix} \rightsquigarrow e^{tA}U = U \begin{pmatrix} e^{t\lambda} & \\ & e^{t\lambda} \end{pmatrix} = \underbrace{\begin{pmatrix} e^{t\lambda}u^{(1)} & e^{t\lambda}u^{(2)} \\ | & | \end{pmatrix}}_{\text{columns span } \mathcal{S}_0}$$

(2b) $\lambda_1 = \lambda_2 =: \lambda$ (A is not diagonalizable)

basis $(u^{(1)}, u^{(2)})$ $\rightsquigarrow U = \begin{pmatrix} | & | \\ u^{(1)} & u^{(2)} \\ | & | \end{pmatrix}$ invertible
 $\swarrow$ eigenvector $\nwarrow$ generalized eigenvector

$$U^{-1}AU = \begin{pmatrix} \lambda & 1 \\ & \lambda \end{pmatrix} = \underbrace{\begin{pmatrix} \lambda & \\ & \lambda \end{pmatrix}}_{\text{Jordan normal form } \mathcal{D}} + \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{\mathcal{N}} \quad \text{with } \mathcal{N}^2 = 0$$

$$U^{-1}e^{tA}U = e^{t(\mathcal{D}+\mathcal{N})} = e^{t\mathcal{D}}e^{t\mathcal{N}} = \begin{pmatrix} e^{t\lambda} & \\ & e^{t\lambda} \end{pmatrix} (\mathbb{1} + t\mathcal{N}) = e^{t\lambda} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

$$\implies e^{tA}U = e^{t\lambda}U \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} = e^{t\lambda} \underbrace{\begin{pmatrix} | & | \\ u^{(1)} & t u^{(1)} + u^{(2)} \\ | & | \end{pmatrix}}_{\text{columns span } \mathcal{S}_0}$$



Ordinary Differential Equations - Part 27

Linear ODE of order 2: (homogeneous + autonomous)

$$\ddot{x} + a_1 \dot{x} + a_0 x = 0, \quad a_0, a_1 \in \mathbb{R}$$

transform to 1st order

$$\begin{aligned} y_1 &= x \\ y_2 &= \dot{x} \end{aligned}, \quad \begin{pmatrix} \dot{y}_1 \\ \dot{y}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix}}_A \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

We have 4 cases for the two-dimensional system: we need **eigenvalues** and **eigenvectors**

zeros of characteristic polynomial

$$p_A(\lambda) = \lambda^2 + a_1 \lambda + a_0$$

non-zero elements of

$$\text{Ker}(A - \lambda \mathbb{1})$$

For eigenvectors:

$$\begin{pmatrix} -\lambda & 1 \\ -a_0 & -a_1 - \lambda \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \leadsto -\lambda w_1 + w_2 = 0$$

eigenvector satisfies: $w_1 \neq 0$

assume without loss of generality: $w_1 = 1 \Rightarrow w_2 = \lambda$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -\lambda & 1 \\ -a_0 & -a_1 - \lambda \end{pmatrix} \begin{pmatrix} 1 \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ -a_0 - a_1 \lambda - \lambda^2 \end{pmatrix} = \begin{pmatrix} 0 \\ -p_A(\lambda) \end{pmatrix}$$

Result: λ eigenvalue $\Leftrightarrow \begin{pmatrix} 1 \\ \lambda \end{pmatrix}$ eigenvector

(eigenspace is 1-dimensional)

4 cases in part 26:

(1a) two different eigenvalues $\in \mathbb{R}$

(1b) two different eigenvalues $\in \mathbb{C} \setminus \mathbb{R}$

~~(2a) only one eigenvalue: A is diagonalizable~~

(2b) only one eigenvalue: A is not diagonalizable

General solutions for $\ddot{x} + a_1 \dot{x} + a_0 x = 0$

don't forget: $y_1 = x$

(1a) $\lambda_1 \neq \lambda_2 \in \mathbb{R}$

solution space spanned by: $e^{t\lambda_1} u^{(1)}, e^{t\lambda_2} u^{(2)}$

first component!

$$t \mapsto e^{t\lambda_1} \text{ and } t \mapsto e^{t\lambda_2}$$

(1b) $\lambda_1 \neq \lambda_2 \in \mathbb{C} \setminus \mathbb{R}$

solution space spanned by

first component!

$$\begin{aligned} \lambda_1 &= \alpha + i\beta \\ \lambda_2 &= \alpha - i\beta \end{aligned}$$

$$e^{t\alpha} (\cos(\beta t) u - \sin(\beta t) v), e^{t\alpha} (\cos(\beta t) v + \sin(\beta t) u)$$

$$t \mapsto e^{t\alpha} \cos(\beta t) \text{ and } t \mapsto e^{t\alpha} \sin(\beta t)$$

(2b) $\lambda_1 = \lambda_2 =: \lambda$

solution space spanned by $e^{t\lambda} u^{(1)}, e^{t\lambda} (t u^{(1)} + u^{(2)})$

first component!

basis $(u^{(1)}, u^{(2)})$
generalized eigenvector
eigenvector

$$t \mapsto e^{t\lambda} \text{ and } t \mapsto t e^{t\lambda}$$



Ordinary Differential Equations - Part 28

(József Maria Hoëné-Wroński)

System of linear differential equations: $\dot{x} = A(t)x + b(t)$, $I \ni t \xrightarrow{\text{continuous}} A(t) \in \mathbb{R}^{n \times n}$
 $I \ni t \xrightarrow{\text{continuous}} b(t) \in \mathbb{R}^n$
interval in $\mathbb{R}$

Solution set: $S = S_0 + \gamma$ $\leftarrow$ particular solution of $\dot{x} = A(t)x + b(t)$
n-dimensional solution space
 $\dot{x} = A(t)x$

Homogeneous part: $\dot{x} = A(t)x$ (for $A(t) = \text{constant}$, matrix exponential helps)

$$S_0 = \text{Span}(\underbrace{\alpha_1, \alpha_2, \dots, \alpha_n}_{\text{basis}}) \quad \text{where } \alpha_j: I \rightarrow \mathbb{R}^n$$

We call the map $U: I \rightarrow \mathbb{R}^{n \times n}$, $t \mapsto \begin{pmatrix} \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \end{pmatrix}$

a fundamental matrix solution.

Definition: For $\dot{x} = A(t)x$ ($I \ni t \xrightarrow{\text{continuous}} A(t) \in \mathbb{R}^{n \times n}$)

and solutions $\alpha_1, \alpha_2, \dots, \alpha_n$, we define $\leftarrow$ matrix solution

$$W(t) = \det \begin{pmatrix} \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \end{pmatrix}$$

and call it the Wronski determinant (or just Wronskian).



Ordinary Differential Equations - Part 29

System of linear ODEs: $\dot{x} = A(t)x$ with $I \ni t \xrightarrow{\text{continuous}} A(t) \in \mathbb{R}^{n \times n}$

interval in $\mathbb{R}$ $\rightarrow$ solutions: $\alpha_1, \alpha_2, \dots, \alpha_n : I \rightarrow \mathbb{R}^n$

matrix solution: $U : I \rightarrow \mathbb{R}^{n \times n}, t \mapsto \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix}$

Wronskian: $W(t) = \det \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix}$

Fact: Assume there is a $t_0 \in I$ with $W(t_0) = 0$

$\Leftrightarrow \alpha_1(t_0), \alpha_2(t_0), \dots, \alpha_n(t_0)$ are linearly dependent in $\mathbb{R}^n$

$\Leftrightarrow$ there are $\lambda_j \in \mathbb{R}$ (not all zero) with: $0 = \sum_{j=1}^n \lambda_j \alpha_j(t_0)$

but zero function solves IVP

$$\begin{cases} \dot{x} = A(t)x \\ x(t_0) = 0 \end{cases}$$

and $t \mapsto \sum_{j=1}^n \lambda_j \alpha_j(t)$ as well!

$\xrightarrow{\text{Picard-Lindelöf}} 0 = \sum_{j=1}^n \lambda_j \alpha_j(t)$ for all $t \in I$

$\Leftrightarrow \alpha_1, \alpha_2, \dots, \alpha_n$ are linearly dependent in $C^1(I, \mathbb{R}^n)$.

Theorem: Consider $\dot{x} = A(t)x$ with $I \ni t \xrightarrow{\text{continuous}} A(t) \in \mathbb{R}^{n \times n}$

interval in $\mathbb{R}$ $\rightarrow$ solutions: $\alpha_1, \alpha_2, \dots, \alpha_n : I \rightarrow \mathbb{R}^n$

Then: (a) The Wronski determinant $W(t)$ satisfies:

$$\dot{W}(t) = \text{tr}(A(t)) W(t) \quad \text{trace of matrix = sum of diagonal}$$

$$(b) \quad W(t) = W(t_0) e^{\int_{t_0}^t \text{tr}(A(s)) ds} \quad (\text{Liouville's formula})$$

$$(c) \quad U : I \rightarrow \mathbb{R}^{n \times n}, t \mapsto \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix}$$

is a fundamental matrix solution $\Leftrightarrow \exists t_0 \in I : W(t_0) \neq 0$

Proof: (a) $\frac{d}{dt} \det \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix} \stackrel{\text{product rule}}{=} \det \begin{pmatrix} \dot{\alpha}_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \\ \alpha_1(t) & \dot{\alpha}_2(t) & \dots & \alpha_n(t) \\ | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \dot{\alpha}_n(t) \\ | & | & & | \end{pmatrix} +$

$$+ \det \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \dot{\alpha}_2(t) & \dots & \alpha_n(t) \\ | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \dot{\alpha}_n(t) \\ | & | & & | \end{pmatrix} + \dots + \det \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \dot{\alpha}_n(t) \\ | & | & & | \end{pmatrix}$$

$$\quad \quad \quad \swarrow \text{expand along second column}$$

Let's simplify the notation: Fix $t \in \mathbb{R}$ and

$$\mathcal{B} := \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & & & & \\ \vdots & & & & \\ b_{n1} & b_{n2} & \dots & & b_{nn} \end{pmatrix}$$

cofactor matrix $C_i : c_{ij} = (-1)^{i+j} \cdot \det \begin{pmatrix} | & | & & | \\ \mathcal{B} & | & & | \\ | & | & & | \end{pmatrix}$ $\begin{matrix} \swarrow \text{jth column deleted} \\ \searrow \text{ith row deleted} \end{matrix}$

$$\mathcal{B} \cdot C_i^T = \det(\mathcal{B}) \cdot \mathbb{1}_n$$

We will use: $(\dot{\alpha}_j(t))_k = (A(t) \alpha_j(t))_k = \sum_{m=1}^n a_{km} b_{mj}$

Hence: $\dot{W}(t) = \sum_{j=1}^n \det \begin{pmatrix} b_{11} & b_{12} & \dots & | & \dots & b_{1n} \\ \vdots & & & | & & \vdots \\ b_{n1} & b_{n2} & \dots & | & \dots & b_{nn} \end{pmatrix} \begin{matrix} \swarrow \text{Laplace expansion} \\ \searrow \text{along jth column} \end{matrix}$

$$= \sum_{j=1}^n \sum_{i=1}^n c_{ij} \sum_{m=1}^n a_{im} b_{mj} = \sum_{j,i,m} b_{mj} (C_i^T)_{ji} a_{im}$$

$$= \sum_{i,m} \left(\sum_j b_{mj} (C_i^T)_{ji} \right) a_{im} = \sum_{i,m} \underbrace{(\mathcal{B} C_i^T)_{mi}}_{= \det(\mathcal{B}) \delta_{mi}} a_{im}$$

$$= \sum_{i=1}^n a_{ii} \cdot \det(\mathcal{B}) = \text{tr}(A(t)) W(t)$$

(b) solution of ODE: $W(t) = W(t_0) e^{\int_{t_0}^t \text{tr}(A(s)) ds}$

(c) $U : t \mapsto \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \alpha_2(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix}$ is not a fundamental matrix solution

$\xLeftrightarrow[\text{Fact}]{\text{Liouville's formula}} \exists t_0 \in I : W(t_0) = 0 \Leftrightarrow \forall t \in I : W(t) = 0$

Hence: $U : t \mapsto \begin{pmatrix} | & | & & | \\ \alpha_1(t) & \dots & \alpha_n(t) \\ | & | & & | \end{pmatrix}$ is a fundamental matrix solution $\Leftrightarrow \exists t_0 \in I : W(t_0) \neq 0$

□



Ordinary Differential Equations - Part 30

System of linear differential equations: $\dot{x} = A(t)x + b(t)$, $I \ni t \xrightarrow{\text{continuous}} A(t) \in \mathbb{R}^{n \times n}$
 $\text{interval} \rightarrow I \ni t \xrightarrow{\text{continuous}} b(t) \in \mathbb{R}^n$

Method of integrating factors for $n=1$: $\dot{x} = a(t)x + b(t)$
 homogeneous part has solution $e^{\int_{t_0}^t a(s) ds} = p(t)$

Find solution of IVP: $\dot{x} = a(t)x + b(t)$, $x(t_0) = x_0$ $p(t_0) = 1$

$$\begin{aligned} \dot{x} - a(t)x &= b(t) \quad \Big| \cdot e^{-\int_{t_0}^t a(s) ds} = \frac{1}{p(t)} = p(t)^{-1} \\ \Leftrightarrow \frac{d}{dt}(p(t)^{-1}x) &= p(t)^{-1}b(t) \quad \Big| \int_{t_0}^t \\ \Leftrightarrow p(t)^{-1}x(t) - x(t_0) &= \int_{t_0}^t p(s)^{-1}b(s) ds \end{aligned}$$

$$\text{solution: } x(t) = p(t) \left(x_0 + \int_{t_0}^t p(s)^{-1} b(s) ds \right)$$

Extension to matrices: (1) product rule: $\frac{d}{dt}(M(t)N(t)) = \frac{d}{dt} \begin{pmatrix} \sum_j m_{1j}(t)n_{j1}(t) & \dots \\ \vdots & \ddots \end{pmatrix}$
 $= \dot{M}(t)N(t) + M(t)\dot{N}(t)$
 $\frac{d}{dt} \sum_j \dot{m}_{1j}(t)n_{j1}(t) = \sum_j (\dot{m}_{1j}(t)n_{j1}(t) + m_{1j}(t)\dot{n}_{j1}(t))$
 (Note: An arrow points from the text "ordinary product rule" to the first term in the second equation.)

(2) derivative of the inverse:

$$\begin{aligned} 0 &= \frac{d}{dt}(M(t)M(t)^{-1}) = \dot{M}(t)M(t)^{-1} + M(t)\frac{d}{dt}(M(t)^{-1}) \\ \Rightarrow \frac{d}{dt}(M(t)^{-1}) &= -M(t)^{-1}\dot{M}(t)M(t)^{-1} \end{aligned}$$

Method of integrating factors for $n \in \mathbb{N}$: $\dot{x} = A(t)x + b(t)$
 fundamental matrix solution $U(t) = \begin{pmatrix} | & & | \\ \alpha_1(t) & \dots & \alpha_n(t) \\ | & & | \end{pmatrix}$

$$U(t) \text{ satisfies: } \dot{U}(t) = \begin{pmatrix} A(t) & | & \alpha_1(t) & \dots & A(t) & | & \alpha_n(t) \end{pmatrix} = A(t) \cdot U(t)$$

Define the principal matrix solution (at t_0): $\Pi(t) = U(t)U(t_0)^{-1}$

$\Pi(t)$ satisfies: $\dot{\Pi}(t) = A(t) \cdot \Pi(t)$, $\Pi(t_0) = \mathbb{1}_n$ (identity matrix)

$$\frac{d}{dt}(\Pi(t)^{-1}) = -\Pi(t)^{-1}\dot{\Pi}(t)\Pi(t)^{-1} = -\Pi(t)^{-1}A(t)$$

Let's start the method: $\dot{x} - A(t)x = b(t) \quad \Big| \Pi(t)^{-1}$

$$\Leftrightarrow \Pi(t)^{-1}\dot{x} - \Pi(t)^{-1}A(t)x = \Pi(t)^{-1}b(t)$$

$$\Leftrightarrow \frac{d}{dt}(\Pi(t)^{-1}x) = \Pi(t)^{-1}b(t) \quad \Big| \int_{t_0}^t$$

$$\Leftrightarrow \Pi(t)^{-1}x(t) - x(t_0) = \int_{t_0}^t \Pi(s)^{-1}b(s) ds$$

solution:
(for IVP)

$$x(t) = \Pi(t) \left(x_0 + \int_{t_0}^t \Pi(s)^{-1} b(s) ds \right)$$

$$\begin{aligned} \dot{x} &= A(t)x + b(t) \\ x(t_0) &= x_0 \end{aligned}$$